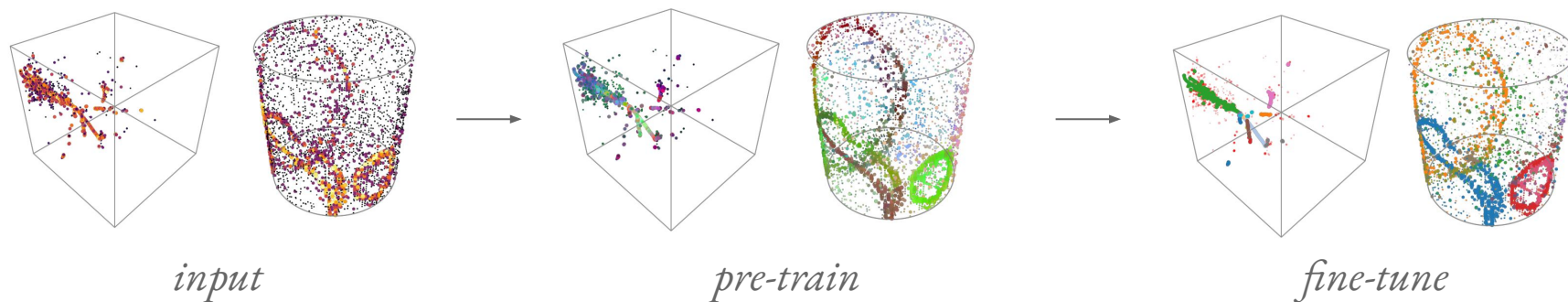


Toward Cross-experiment Pretraining of Point Cloud Foundation Models for Neutrino Detectors



Samuel Young (Stanford)

Kazuhiro Terao (SLAC)

NPML 2026 // based on [arXiv:2512.01324](https://arxiv.org/abs/2512.01324)



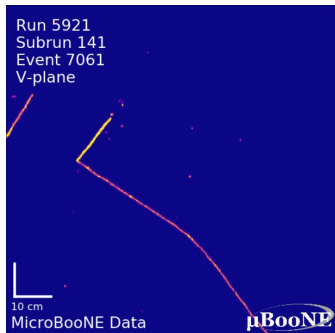
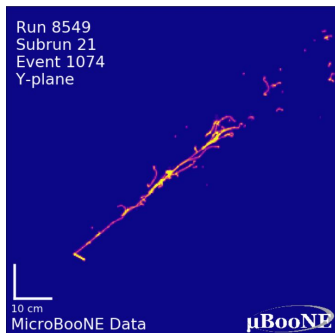
NATIONAL
ACCELERATOR
LABORATORY

Outline

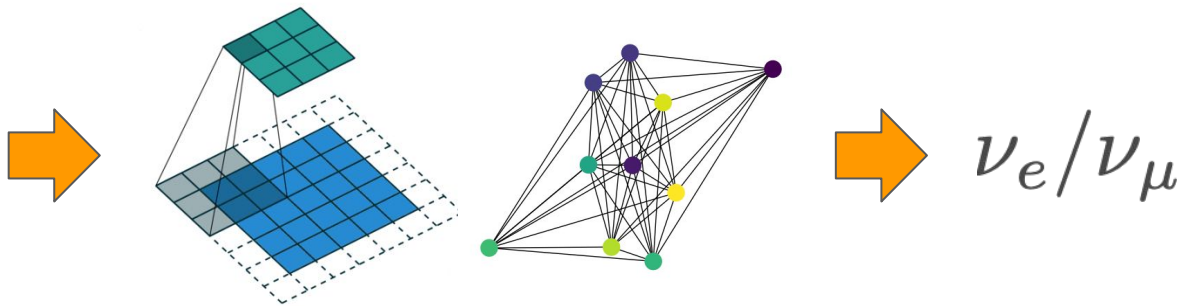
1. Foundation models for sensory data: what & why
2. Panda SSL in LArTPC
3. Panda SSL in Water Cherenkov
4. Lessons learned and challenges for the future

Deep learning in neutrino physics

Common approaches: deep neural network(s) trained using simulated datasets with “labels” via “supervised learning”



Deep Neural Nets



“100 analyses, 100 AI models” (task-specific)

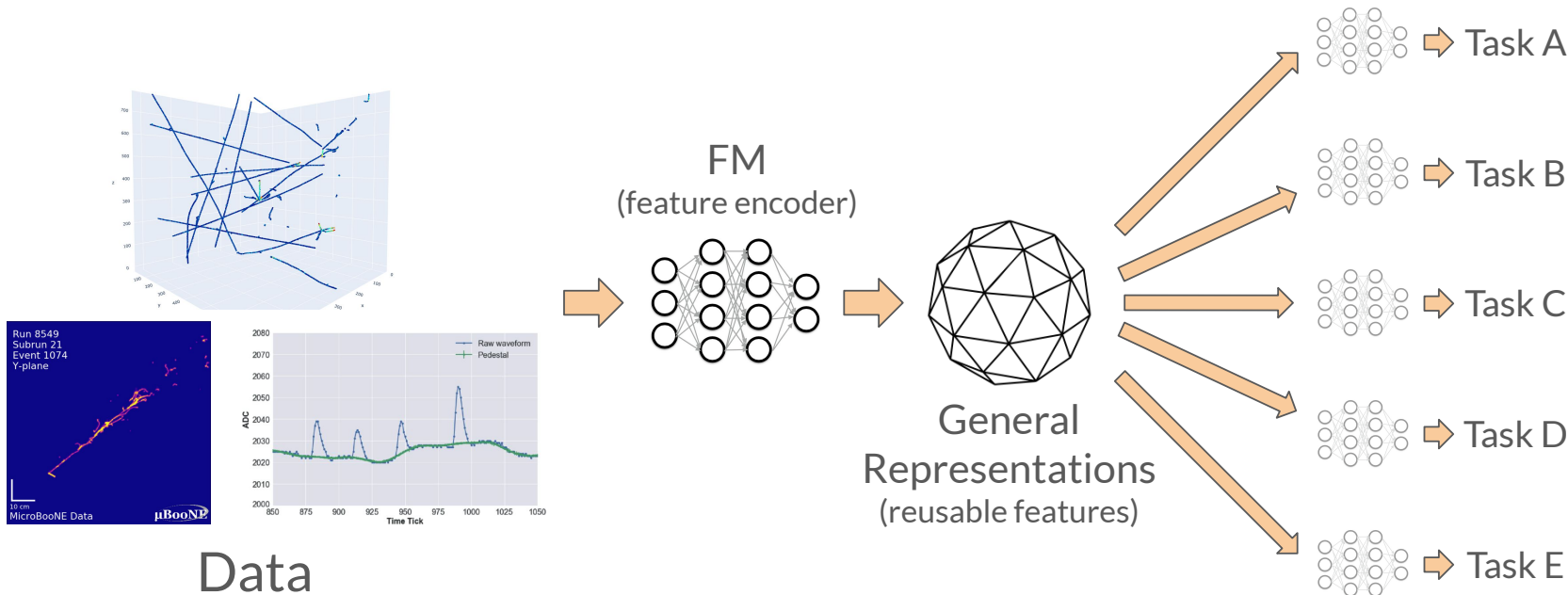
Poor reusability, cost for R&D/maintenance

Vulnerable against data-shift

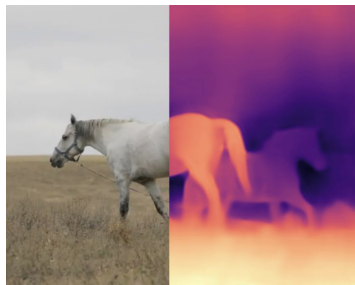
High resolution + sensor-level $\rightarrow$ hard to model.

Foundation models = 2 training phases

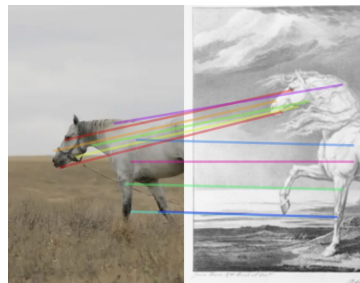
1. **Pre-training** (Representation Learning) – computationally expensive
2. **Fine-tuning** (Adaptation) – computationally inexpensive (relatively speaking...)



Foundation models are generalists

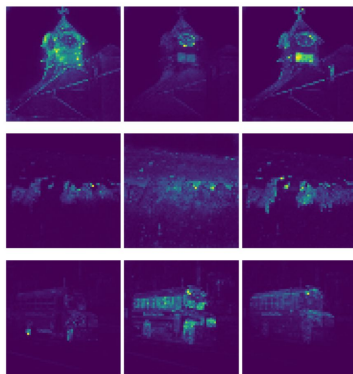


Monocular depth estimation
[1]



Point Correspondence [2]

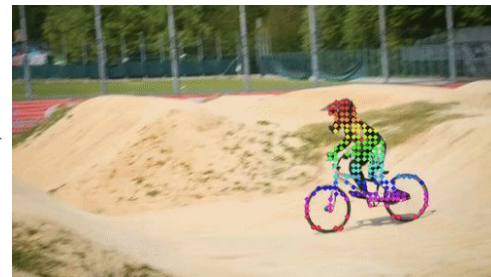
self-supervised vision models
are foundation models



DINO (SSL) [2]



Segmentation [3]



Video Tracking [4]

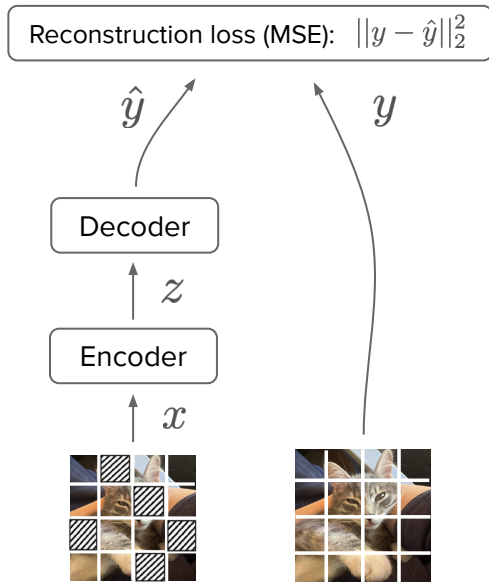
Techniques in self-supervised learning

learning data representations without label supervision by comparing altered/partial views of the same input.

Reconstruction-based methods

[MAE](#), [PoLAR-MAE](#), [FM4NPP](#)

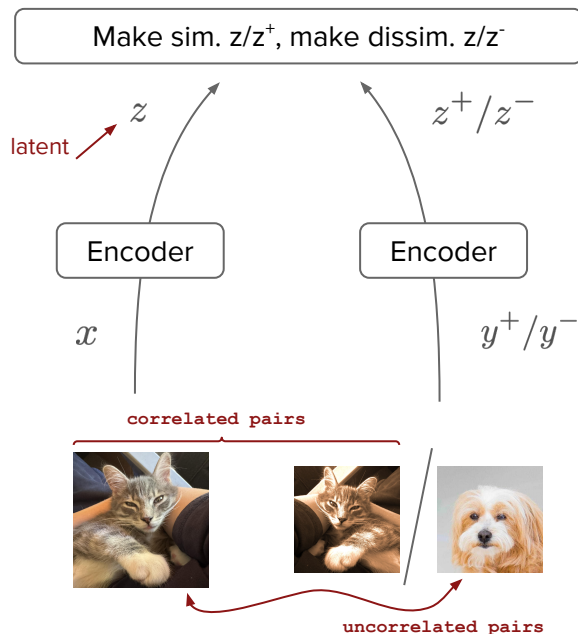
works in pixel space



Contrastive SSL

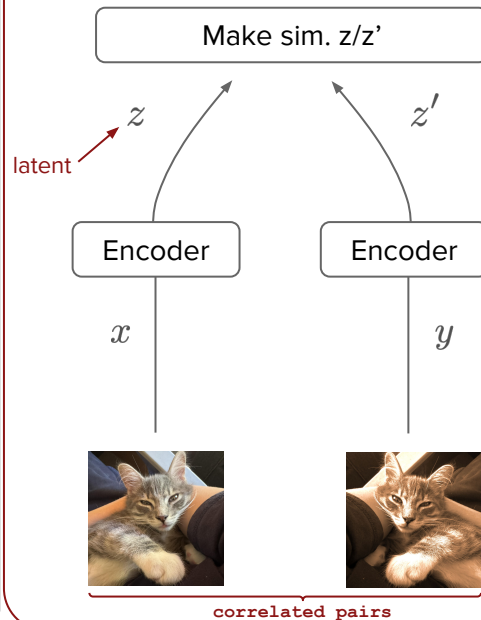
[SimCLR](#), [MoCo](#)

works in latent space



Non-Contrastive SSL

[BYOL](#), [SimSiam](#), [DINO](#), [VICReg](#), [Sonata](#)



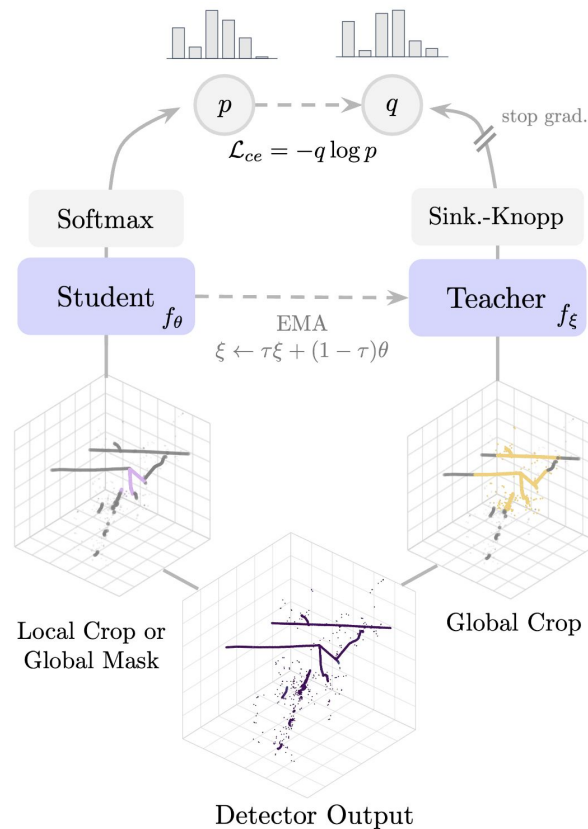
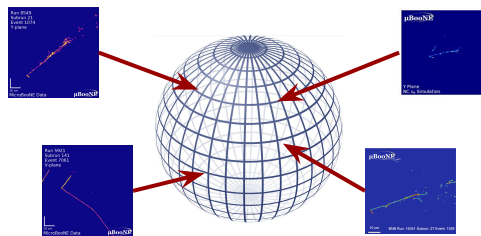
Panda: Self-distillation and hierarchy

instead of **reconstructing masked portions of image directly**,

let's **predict where they would end up on a unit sphere** (i.e., classify them),

by enforcing **consistency between global and local views** of the same image under **strong augmentations**.

this is **non-contrastive SSL**.

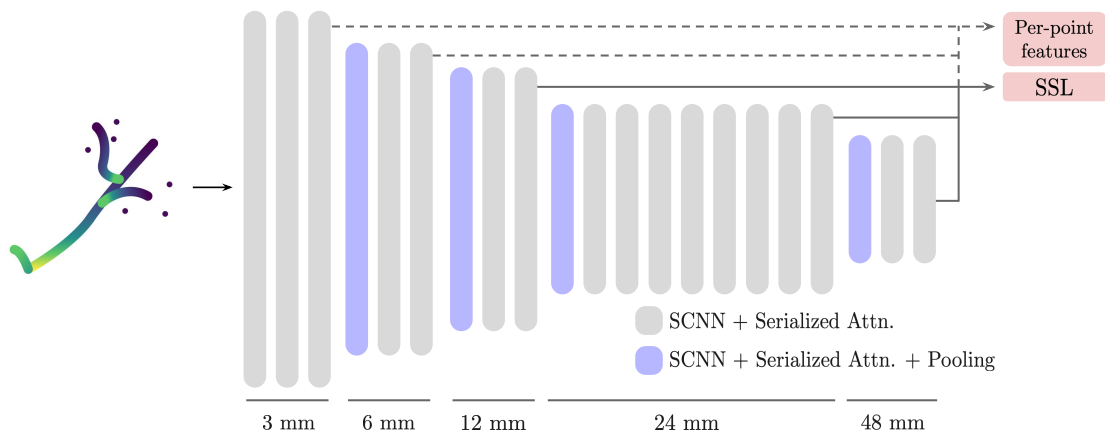
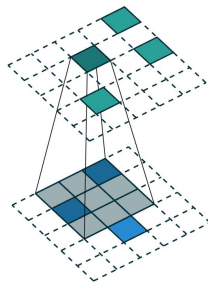


Panda: Self-distillation and hierarchy

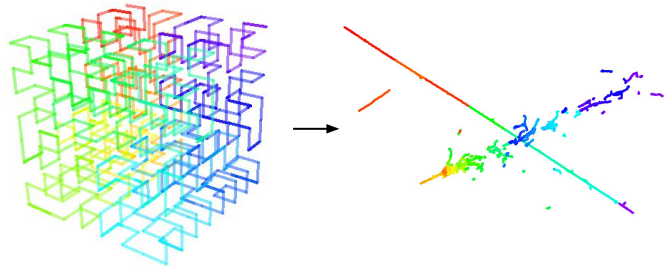
Point Transformer V3 ([CVPR '24](#))

- Efficient serialization for approximate nearest neighbor (no kNN!)
- Local patch hierarchical attention
 - Pooling 5x, 48 to 512 features, patch size 256; 90M params
- Result: more expressive than Sparse UResNet, still scalable (compute scales linearly wrt # points)
- Dataset: [PILArNet-M](#) = 1M 3D point clouds (10k~100k/image)

SCNN: Sparse CNN



Approximate nearest neighbor via locality-sensitive hashing (Hilbert curve, Z-curve)



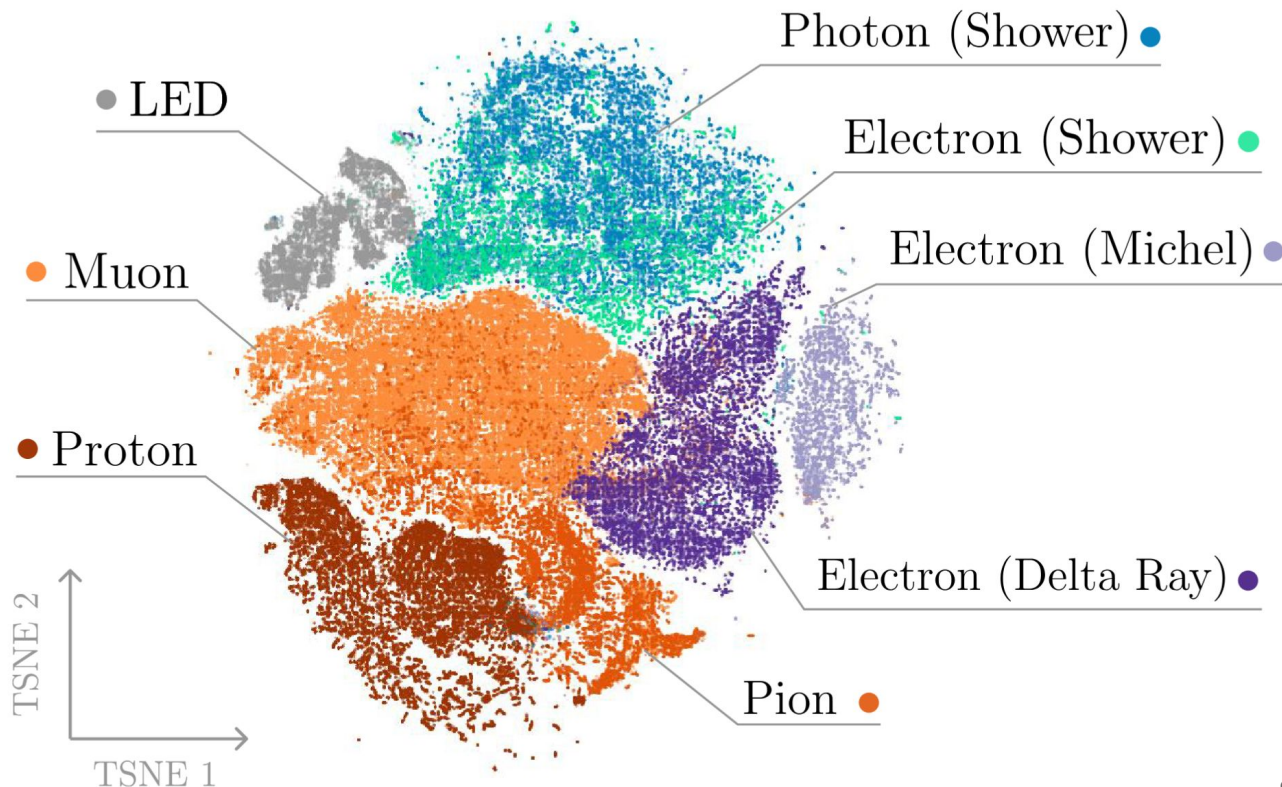
Serialize → cut into non-overlapping patches → windowed attention

(Note: muon length $O(1\text{ m}) \Rightarrow \sim 20$ coarse voxels / muon)

Learned representations

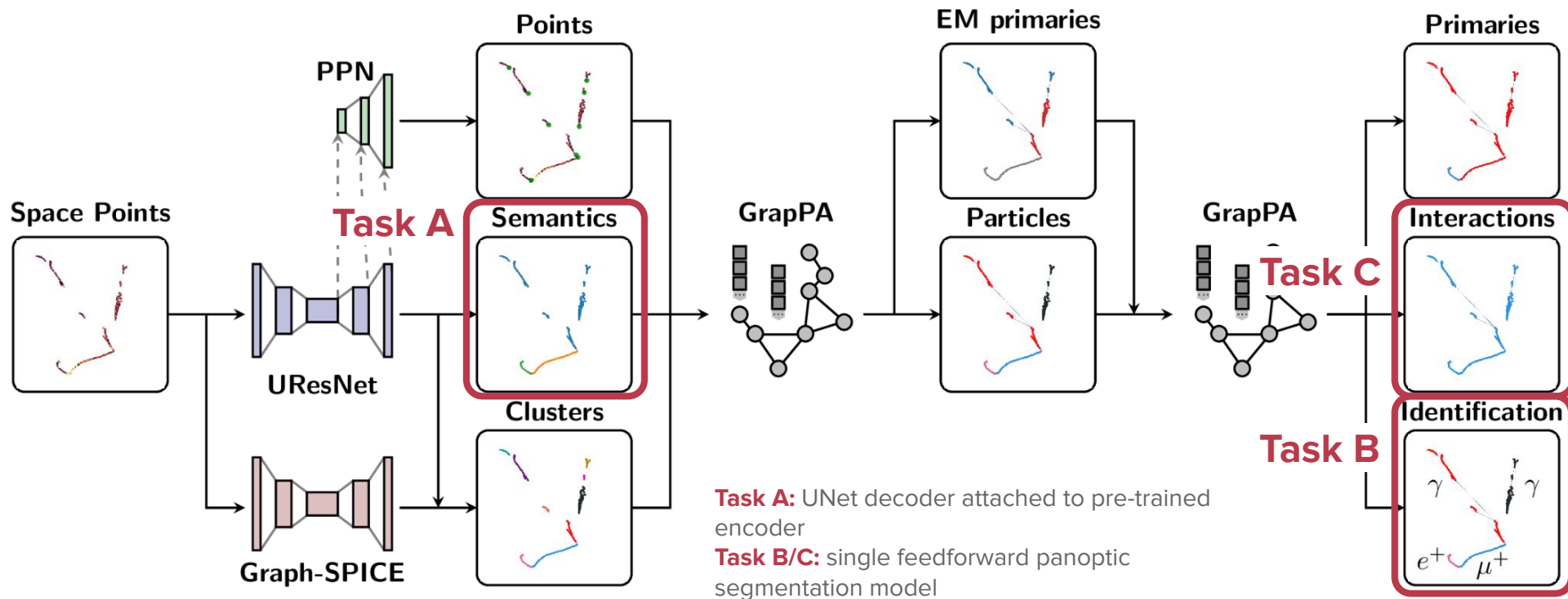
DSet: [PILArNet-M](#)

- 1M simulated MPVMPPR (particle bomb) events
- 768^3 image
- $\sim 2\text{-}30\text{k}$ points/event
- $\sim 4 \pm 3$ interactions/img
- $\sim 11 \pm 8$ particles/img

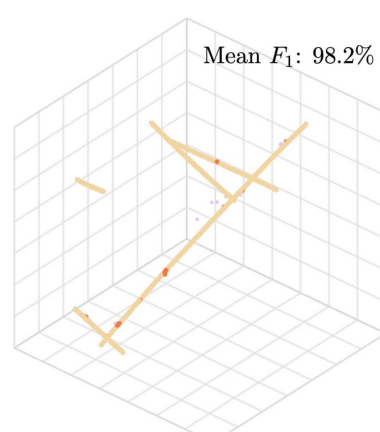
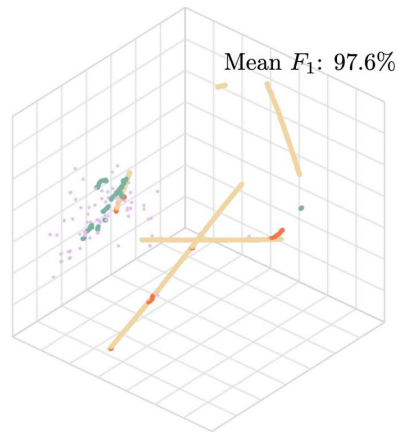
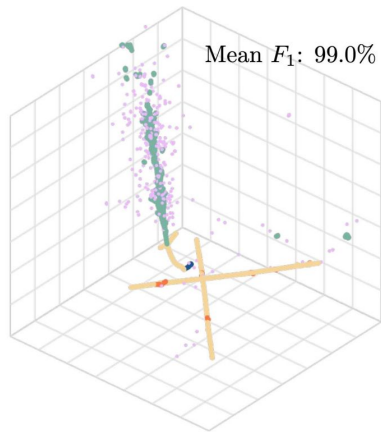
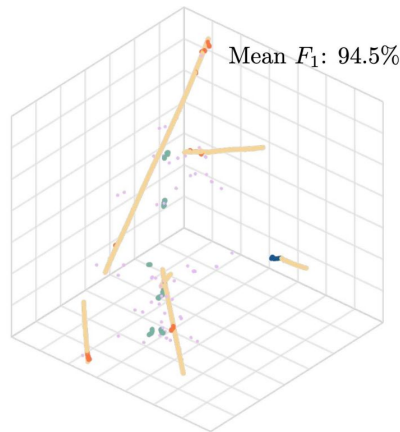


Three tasks

semantic segmentation (A), particle identification (B), interaction identification (C)

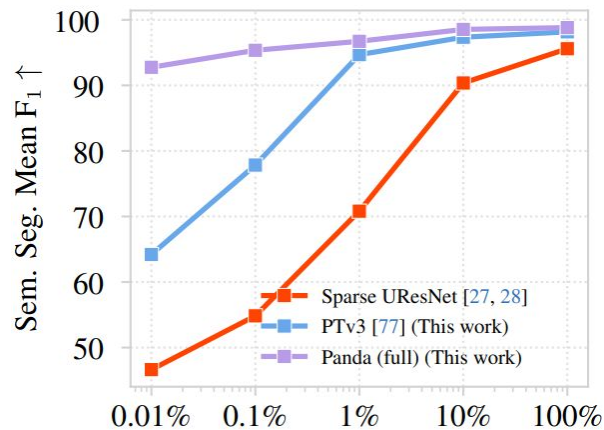


Task A: Semantic Segmentation

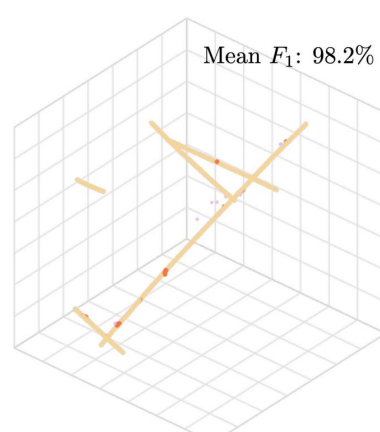
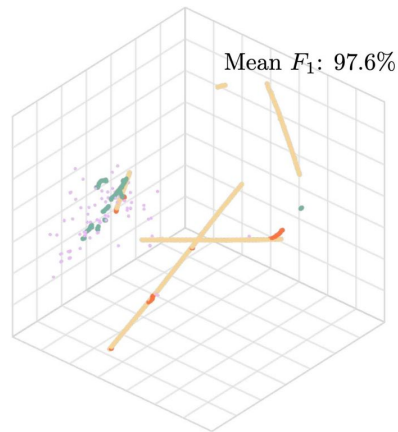
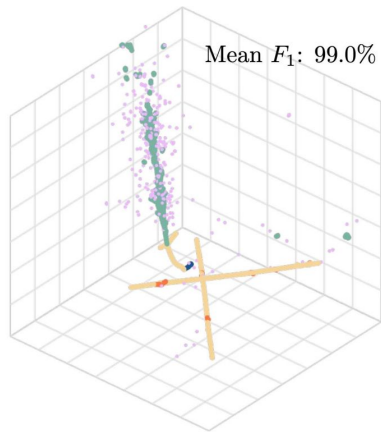
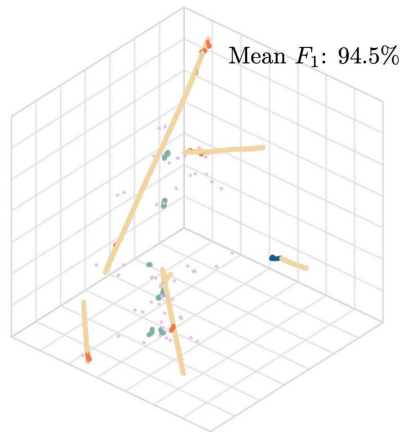


Semantic Segmentation Method / $K\%$	PT $_{K\%}$ + FT $_{K\%}$					PT $_{100\%}$ + FT $_{K\%}$				
	0.01%	0.1%	1%	10%	100%	0.01%	0.1%	1%	10%	100%
<i>Supervised</i>										
○ UResNet [27, 28]	46.6	54.8	70.8	90.4	95.6	46.6	54.8	70.8	90.4	95.6
● PTv3 [77]	64.2	77.8	94.7	97.3	98.2	64.2	77.8	94.7	97.3	98.2
<i>Self-supervised</i>										
● Panda (lin.)	79.1	90.1	93.2	93.7	93.9	92.2	93.6	93.8	93.9	93.9
● Panda (dec.)	83.5	92.8	95.9	97.3	98.2	92.6	95.2	96.2	97.8	98.2
● Panda (full)	85.2	93.7	96.4	97.7	98.8	92.8	95.4	96.7	98.5	98.8

○ Prev. SOTA ● This work

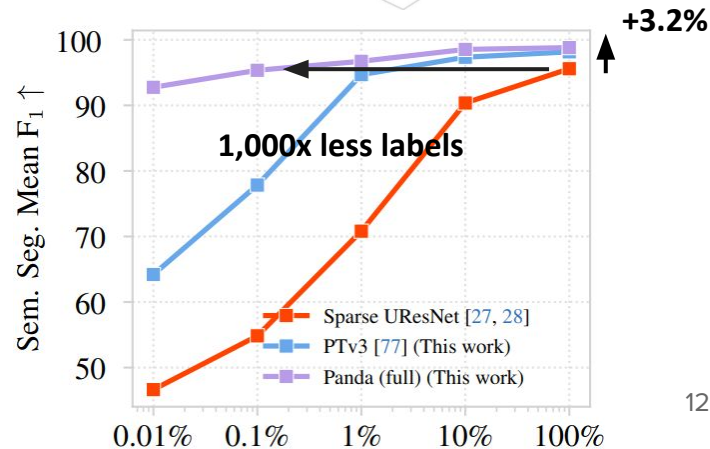


Task A: Semantic Segmentation

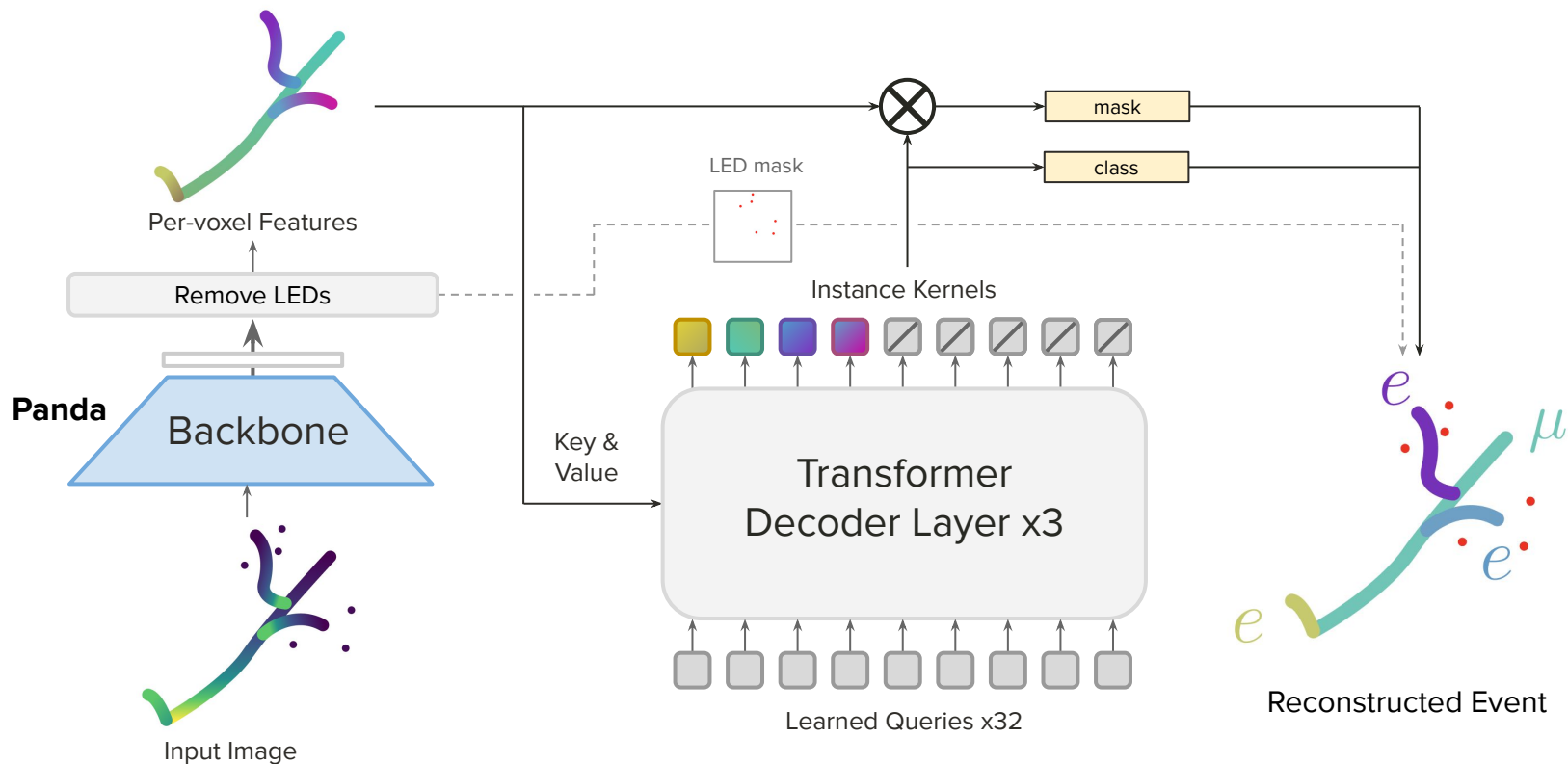


Semantic Segmentation	PT $_K$ % + FT $_K$ %					PT $_{100\%}$ + FT $_K$ %				
	0.01%	0.1%	1%	10%	100%	0.01%	0.1%	1%	10%	100%
<i>Supervised</i>										
○ UResNet [27, 28]	46.6	54.8	70.8	90.4	95.6	46.6	54.8	70.8	90.4	95.6
● PTv3 [77]	64.2	77.8	94.7	97.3	98.2	64.2	77.8	94.7	97.3	98.2
<i>Self-supervised</i>										
● Panda (lin.)	79.1	90.1	93.2	93.7	93.9	92.2	93.6	93.8	93.9	93.9
● Panda (dec.)	83.5	92.8	95.9	97.3	98.2	92.6	95.2	96.2	97.8	98.2
● Panda (full)	85.2	93.7	96.4	97.7	98.8	92.8	95.4	96.7	98.5	98.8

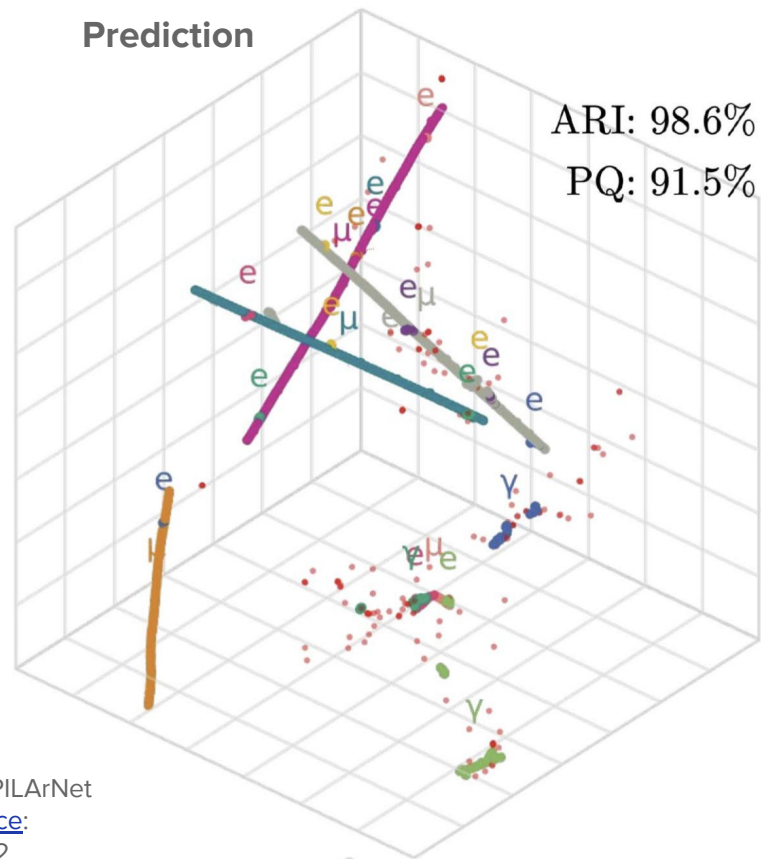
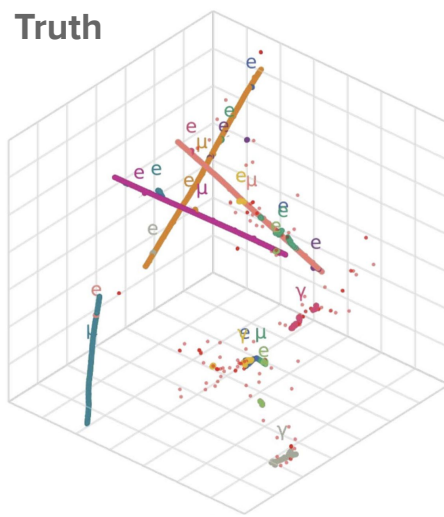
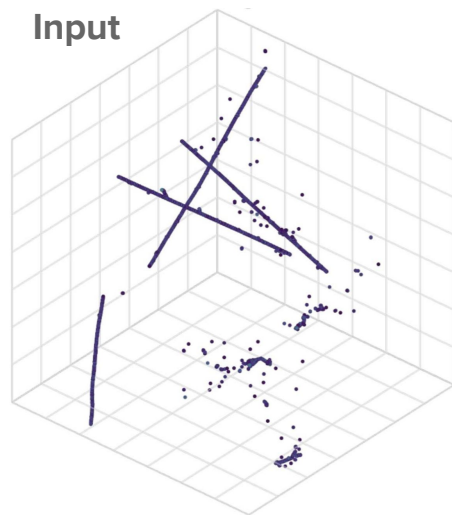
○ Prev. SOTA ● This work



Instance Segmentation: separating particles from one another



Task B: Particle Identification

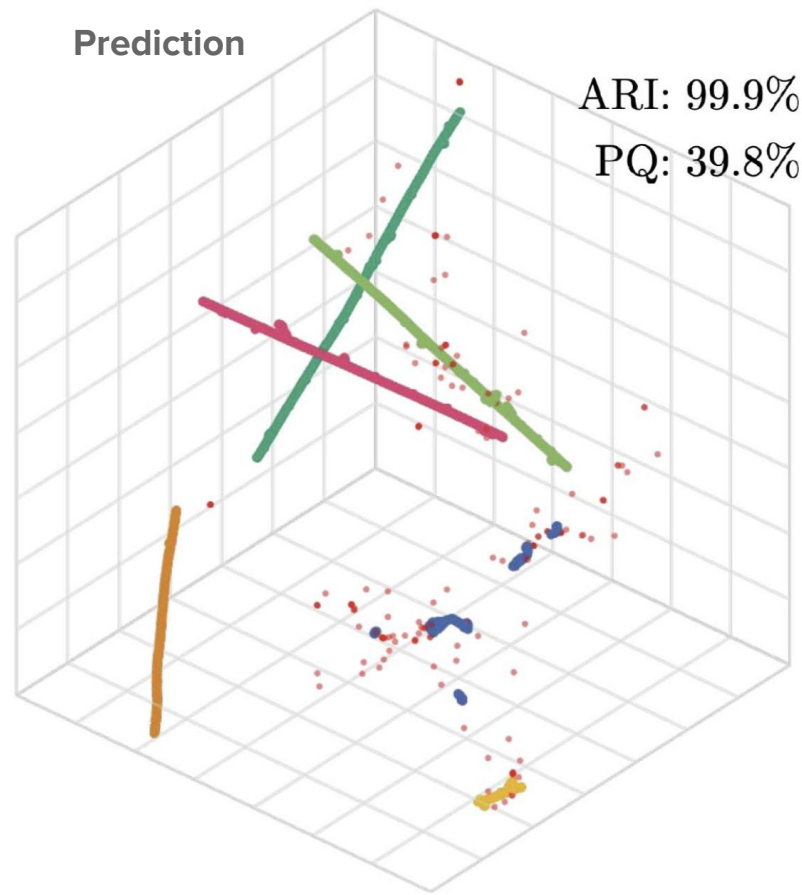
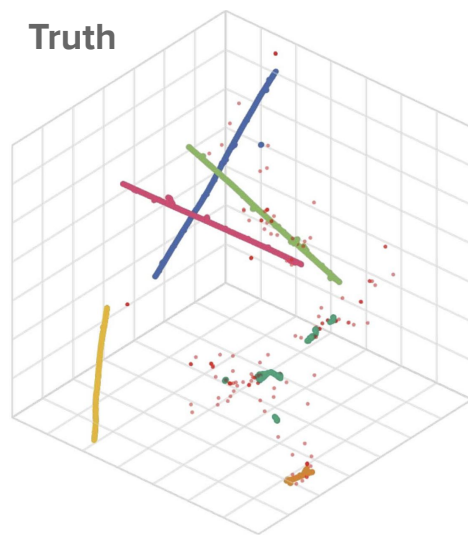
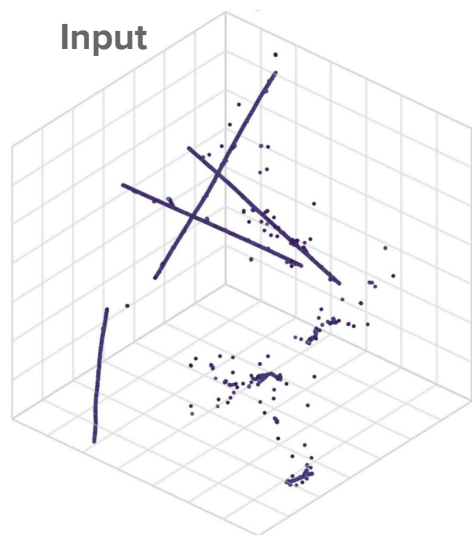


Instance Seg.	Param.	Interaction Id.		Particle Identification							
				PQ	ARI	γ	e	μ	π	p	
<i>Supervised</i>											
• PTv3 [77]	95.1 M	96.3	93.7	86.9	96.6	93.5	94.9	98.6	94.2	98.1	
<i>Self-supervised</i>											
• Panda (dec.)	4.4 M	96.6	94.5	89.5	97.3	96.2	96.3	99.1	95.7	98.4	
• Panda (full)	95.1 M	97.6	94.9	92.5	98.0	98.4	97.2	99.3	96.0	98.6	

• This work

[SPINE](#) PILArNet
reference:
ARI 98.2

Task C: Interaction Identification

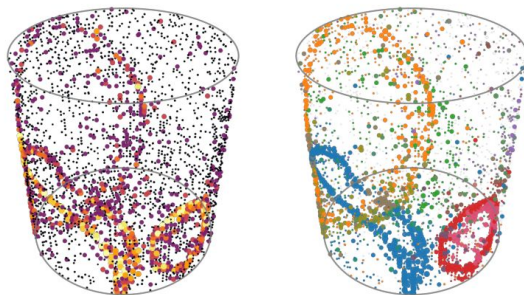


Instance Seg.	Param.	Interaction Id.		Particle Identification						
Method		PQ	ARI	PQ	ARI	γ	e	μ	π	p
<i>Supervised</i>										
• PTV3 [77]	95.1 M	96.3	93.7	86.9	96.6	93.5	94.9	98.6	94.2	98.1
<i>Self-supervised</i>										
• Panda (dec.)	4.4 M	96.6	94.5	89.5	97.3	96.2	96.3	99.1	95.7	98.4
• Panda (full)	95.1 M	97.6	94.9	92.5	98.0	98.4	97.2	99.3	96.0	98.6

• This work

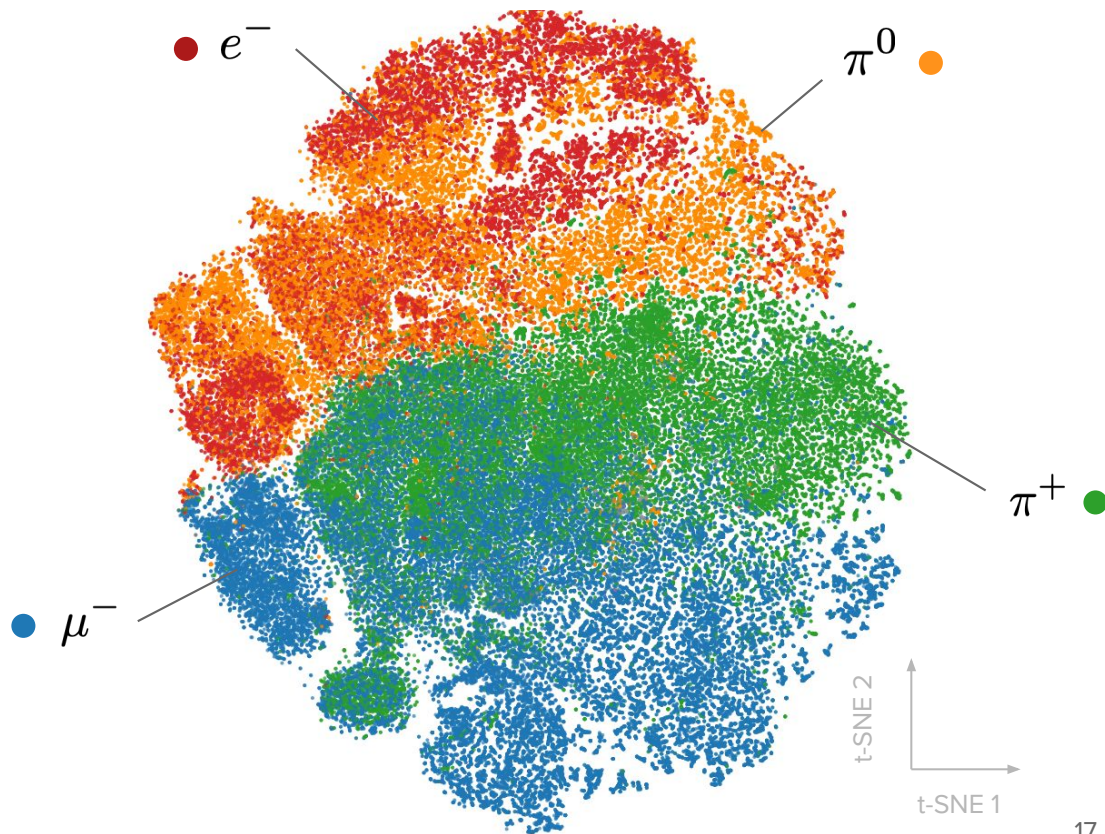
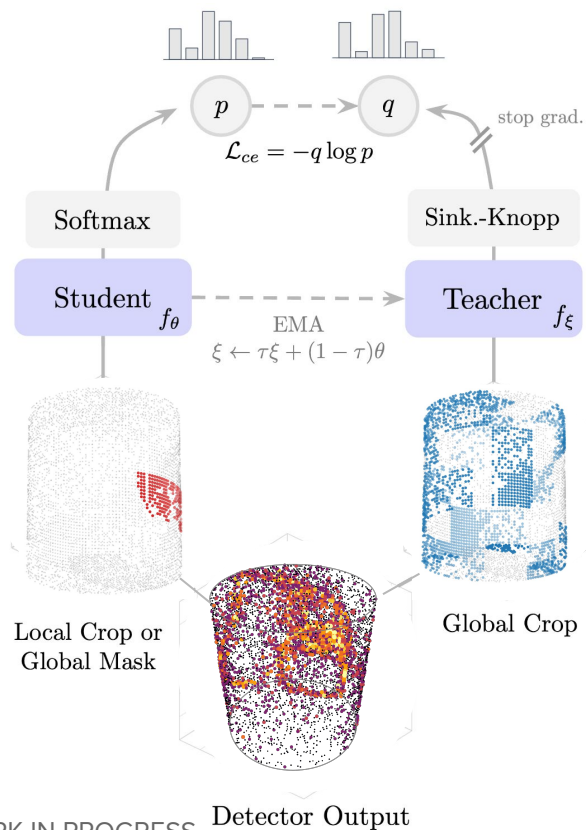
WAND: “Water-cherenkov Annotated Neutrino Dataset”

- New open dataset! Full WCh detector sim with:
 - (GENIE +) Geant4 + [LUCiD](#)
 - Incl.:
 - Relevant scattering, absorption, etc. all wavelength dependent where applicable
 - PMT sensor response (TTS, QE, Aol acceptance, etc.)
- SuperK-like geometry: (~11k PMTs, 16.9m cylinder)
- Single particle (mu, e, pi+, pi0) as well as multi-ring and GENIE nue/numu
- Isotropic particle guns, uniform 1-2000 MeV
- For this work, ~120k total events, ~10k from each config
- See [Cesar's talk](#) tomorrow for in-depth information.



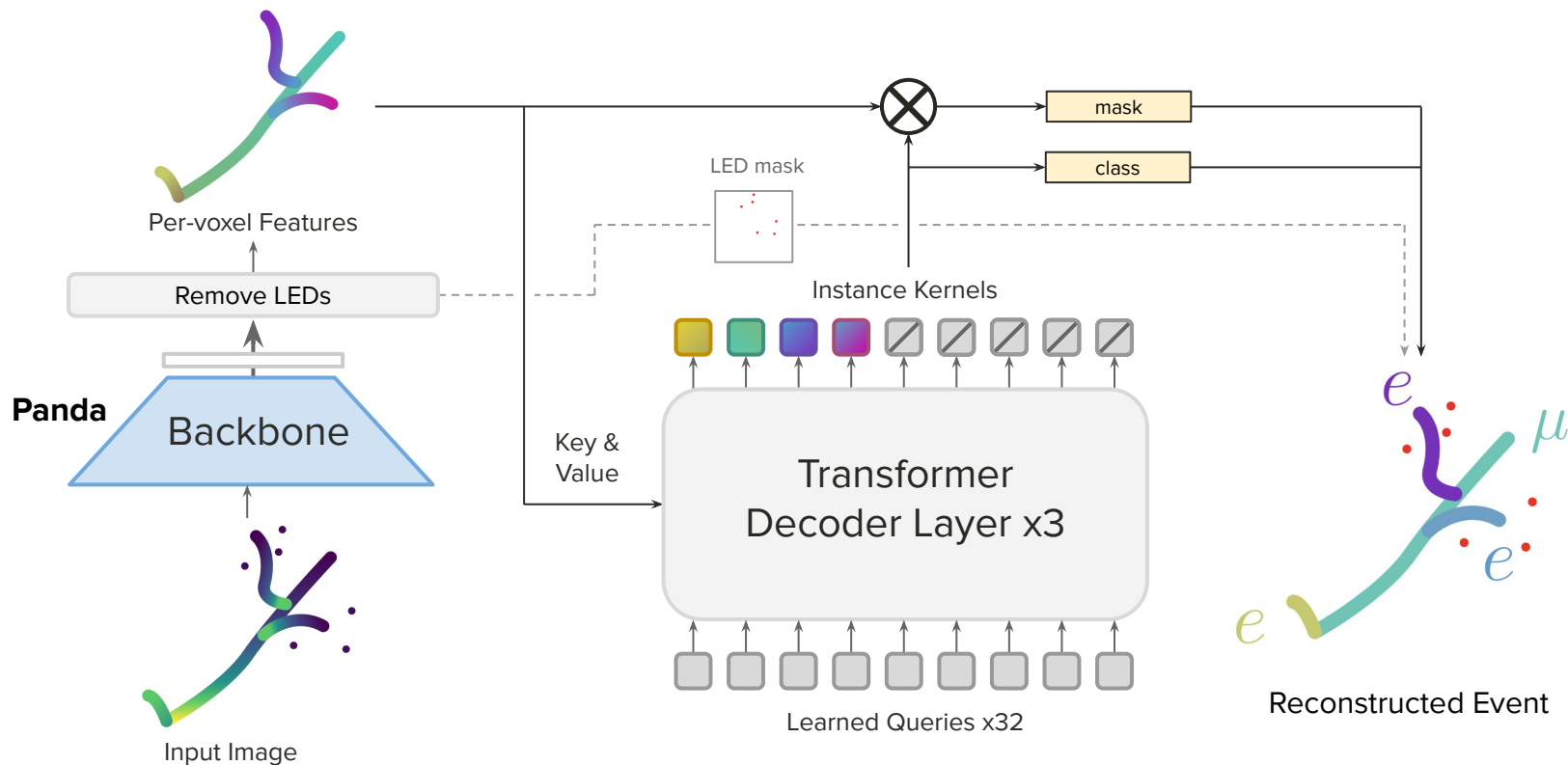
Particle content
μ^-
π^+
e^-
π^-
$\pi^0 (\rightarrow \gamma\gamma)$
e^- (low-energy)
μ^-, π^+
e^-, π^+
$e^-, \pi^0 (e+\gamma\gamma)$
μ^-, π^+, π^0
μ^-, π^+, π^-
e^-, π^+, π^0
ν_μ interaction (CC/NC)
ν_e interaction (CC/NC)
$\mu^- + \pi^+$ gun $\oplus \nu_\mu$
$\mu^- \oplus \pi^+$ (2 vtx)
$\nu_\mu \oplus \nu_e$
2 bombs, 1-5 particles each

Learned representations

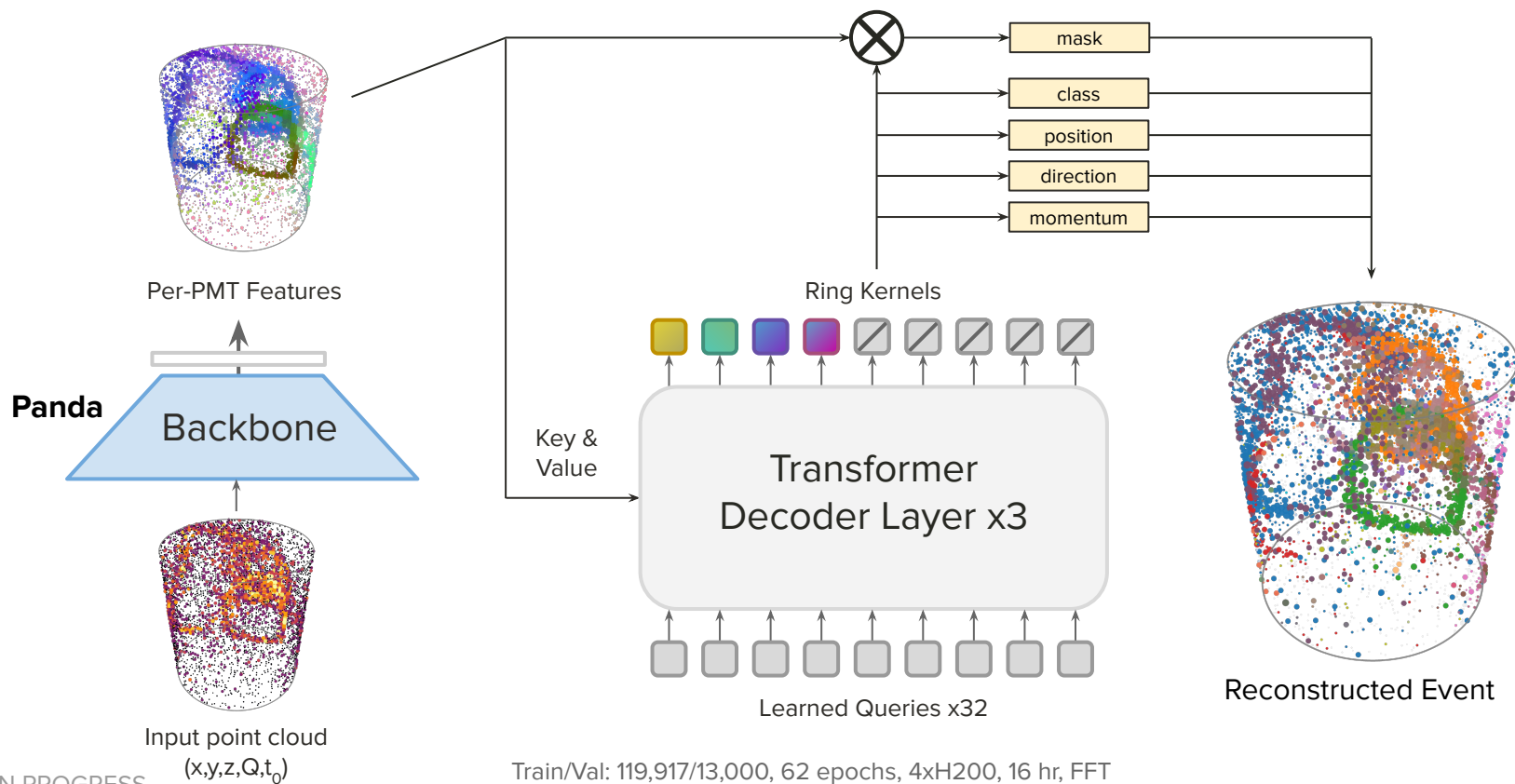


Pre-train: 119,917 events, 50 epochs, 4xH200, 36 hr

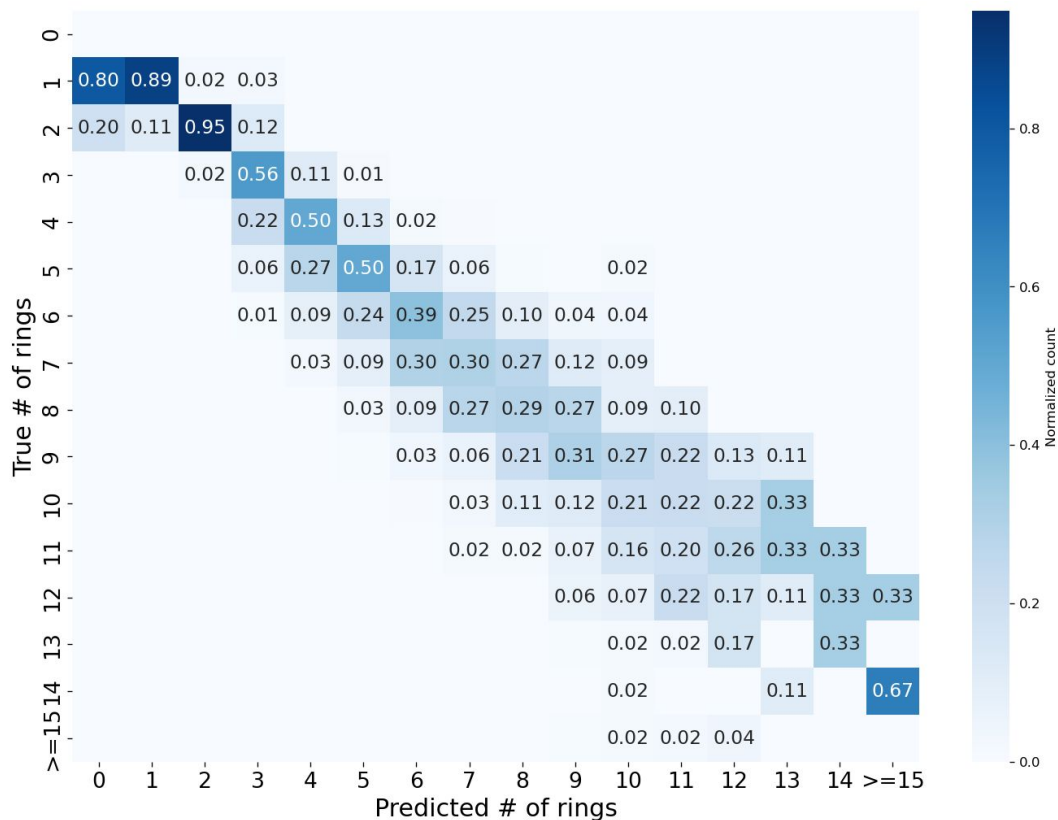
Instance Segmentation: separating particles from one another



Instance Segmentation: separating ~~particles~~ **rings** from one another



General ring counting



Note:

- Using entire validation set
- Ring = any particle that has recorded PMT hits

True Number of Rings	fiTQun Reconstruction		
	1R	2R	$\geq 3R$
True 1R	95.0%	4.64%	0.41%
True 2R	27.8%	66.7%	5.56%
True $\geq 3R$	7.04%	25.5%	67.5%

FiTQun on fully contained atmospheric
(not apples to apples) ([arXiv:1901.03230](https://arxiv.org/abs/1901.03230))

Single particle reconstruction

Cuts:

dist $\rightarrow$ wall > 2 m

$E_{\text{vis}} < 1330$ MeV

Fully contained

99.9% e- efficiency @ 99.9% mu- rejection

85.5% e- efficiency @ 95% pi0 rejection

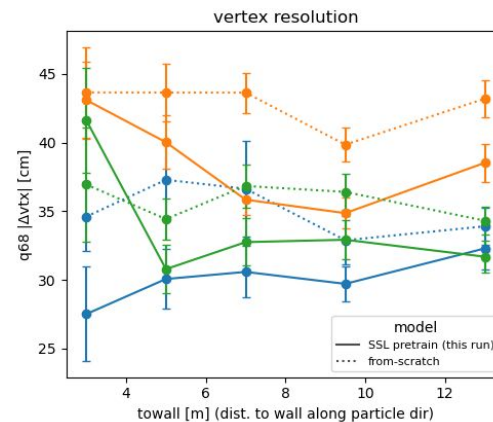
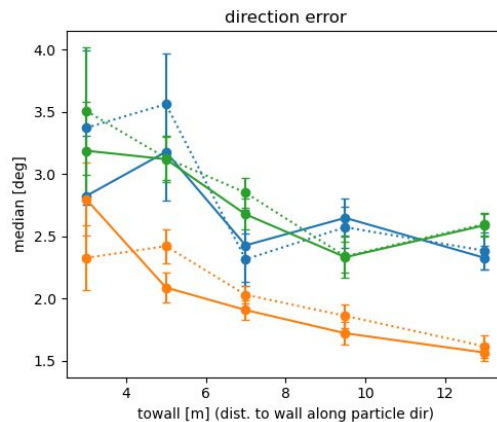
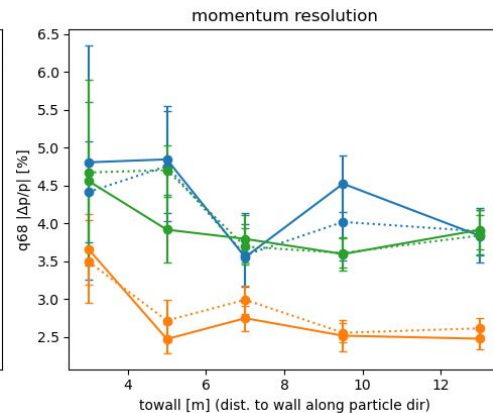
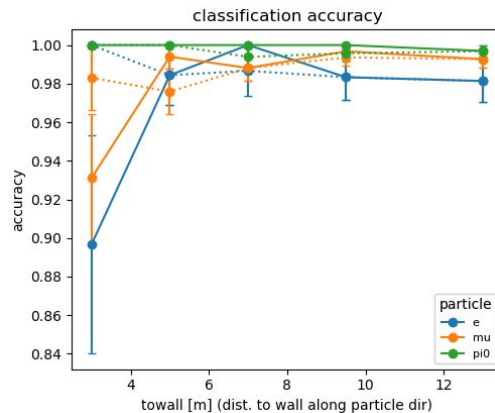
e

mu

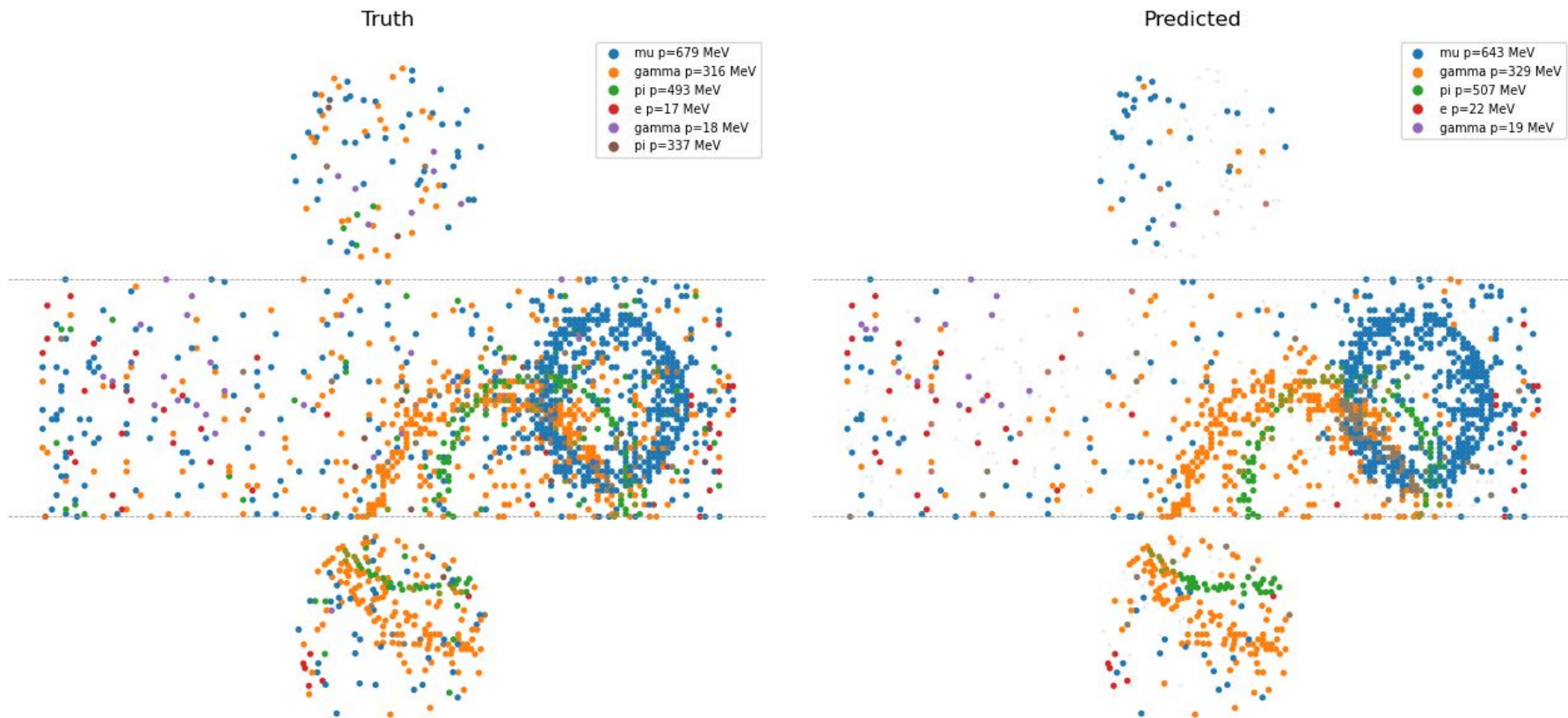
pi0

— SSL pretrain + finefine

- - from-scratch

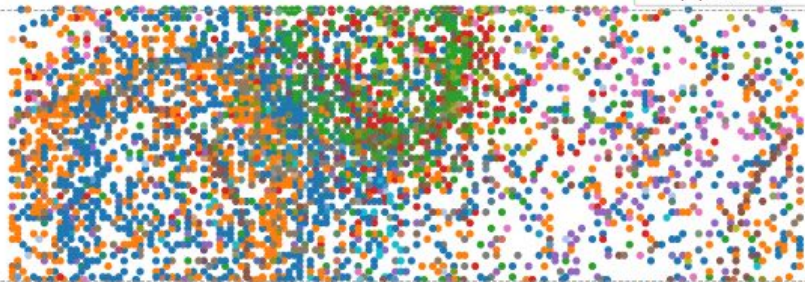
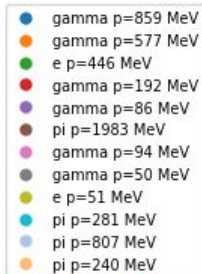


Multi-ring reconstruction

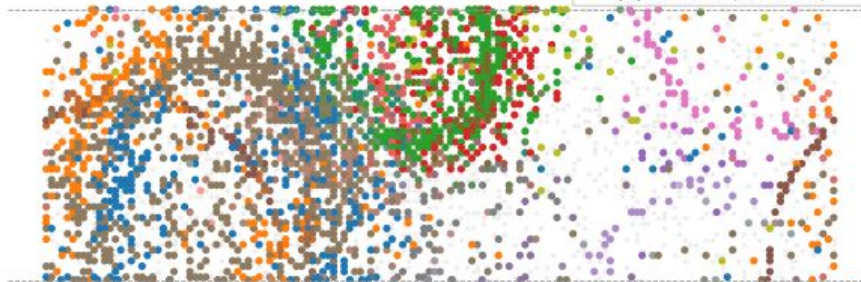
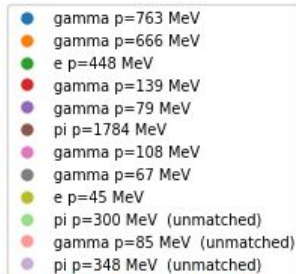


Multi-ring reconstruction

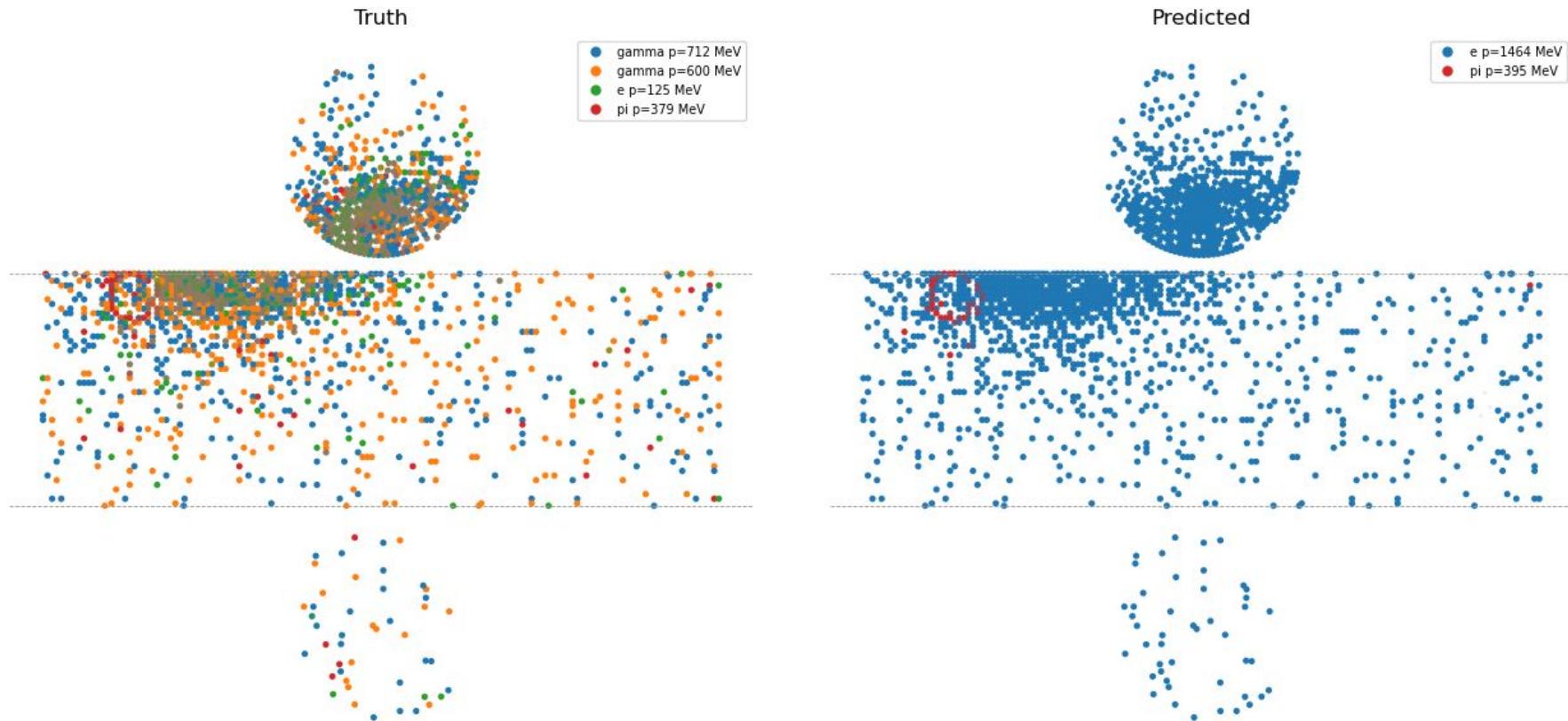
Truth



Predicted



Multi-ring reconstruction



π^0 mass reconstruction using 1,000 $e^- + \pi^0$ events

Cut (roughly following [arXiv:1901.03230](https://arxiv.org/abs/1901.03230))

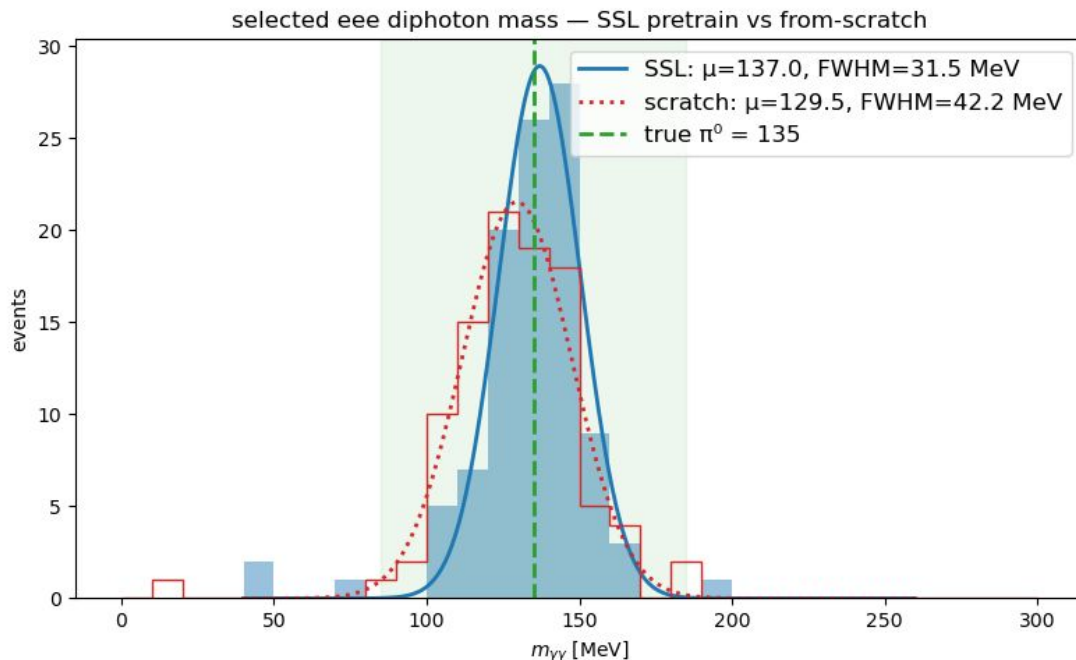
- Min points in ring: 5
- High confidence (> 0.9) PID only
- $\text{dist} \rightarrow \text{wall} > 2$ m
- $\text{dist} \rightarrow \text{wall}$ along particle dir > 2 m
- $E_{\text{vis}} < 1330$ MeV
- 3 predicted rings only
- All rings e-like {e, gamma}

(1000 $\rightarrow$ ~ 100 events)

Reconstruction:

- PT+FT: 137.0 $\pm$ 31.5
- Scratch: 129.5 $\pm$ 42.2

Note: not $\text{CCv}_e 1\pi^0$, so e and π^0 are uncorrelated

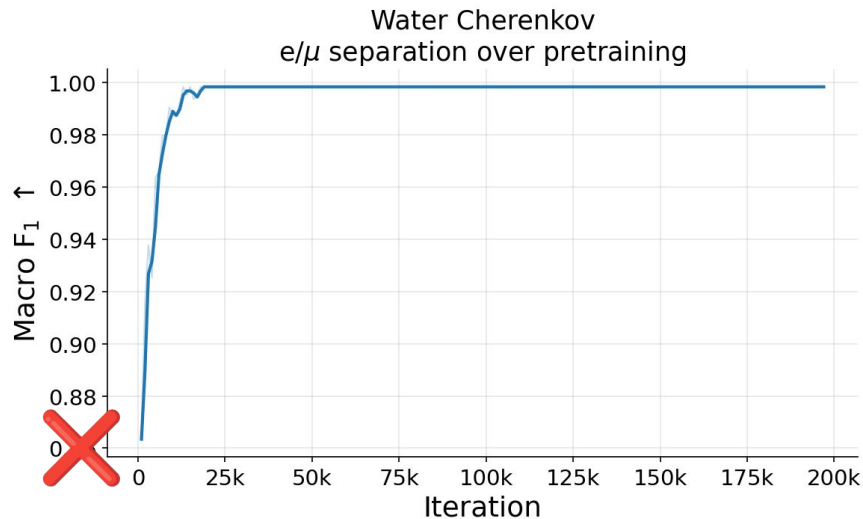
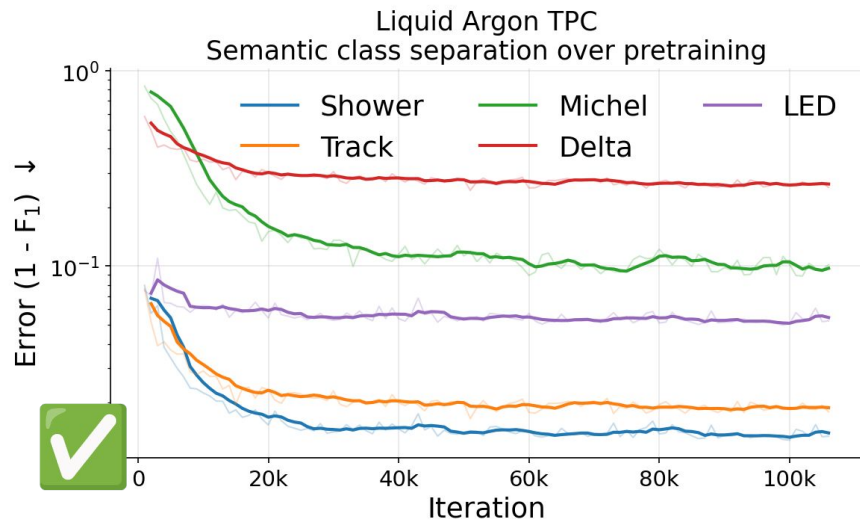


$$m_{\gamma\gamma} = \sqrt{2E_1E_2(1 - \cos\theta_{12})}$$

Lessons learned, and challenges ahead

1. Think of cheap, but difficult, probes that you can use to see how well SSL is learning over the course of training.

Getting SSL to work is extremely difficult and computationally time-intensive. However in these methods I've found that you can see whether it's "working" quite early in training if you run linear probes on weight checkpoints.



Lessons learned, and challenges ahead

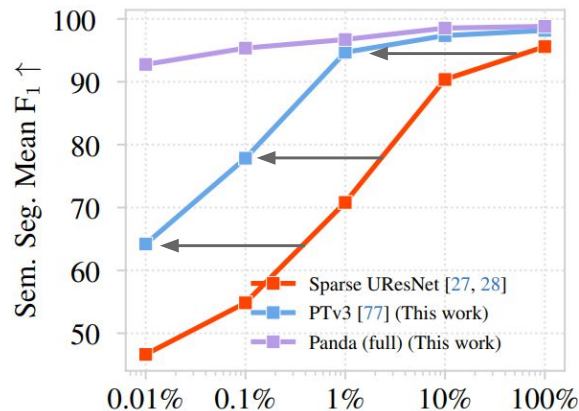
2. Data augmentations can contribute to an effective increase in dataset size by 1 OOM, generally ([arXiv:2106.10270](https://arxiv.org/abs/2106.10270))

- Unphysical augmentations (e.g., random rotations in a LArTPC) can be extremely beneficial for pattern recognition tasks. Think of it as jointly learning the “superset” of transformations, including physical + non-physical.
- For places where you need exact physics (e.g., deriving space charge effects), small fine-tuning models should (imo) be sufficient (but this is not proven).

Difference between **orange** and **blue**

=

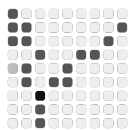
Improved architecture + data augmentation



Lessons learned, and challenges ahead

3. FM R&D requires designing systems that are general, extensible, and can scale.

Consider using **pimm** if starting out!



particle-imaging-models

Train, fine-tune, eval scripts of foundation models for neutrino physics

[github] <https://github.com/deeplearnphysics/particle-imaging-models>

Some features:

- Apptainer/Docker images
- Easy logging to Weights and Biases (wandb)
- Automatic mixed precision (AMP) training.
- Multi-node training using `torchrun`
- DDP for multi-GPU and multi-node training, with (far) future support for FSDP2.
- Fully deterministic checkpointing: start and stop runs as if it never stopped.
- Auto-export and loading models from HuggingFace.
- Validated on [SLAC S3DF](#) and [NERSC Perlmutter](#) using site-specific configurations.
- Point cloud only (for now—1D & 2D in the works)
- NVIDIA only

Local

```
> pimm launch \  
>   --resources.nproc_per_node 2 \  
>   --config pretrain/panda.py
```

SLURM Cluster

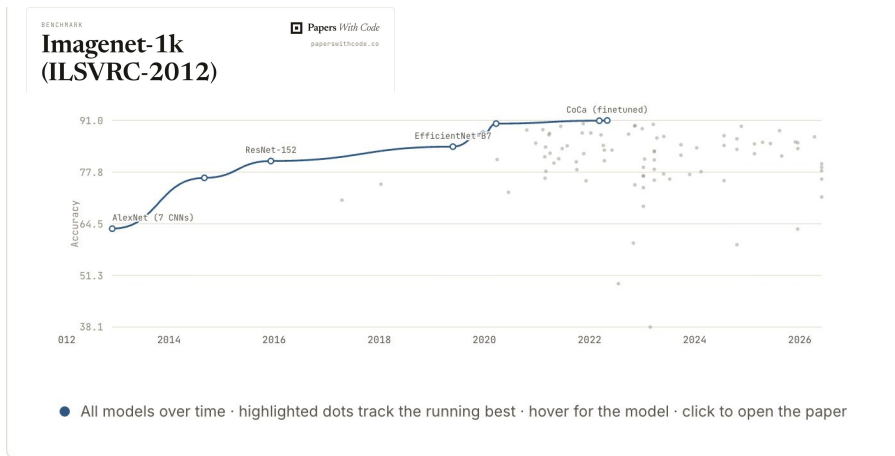
```
> pimm submit \  
>   --site nersc \  
>   --resources.nnodes 2 \  
>   --resources.nproc_per_node 4 \  
>   --resources.time 24:00:00 \  
>   --config pretrain/panda.py
```

Lessons learned, and challenges ahead

1. Common benchmarks and reproducibility

- Deep learning has proliferated because of benchmarks (ImageNet, ScanNet, S3DIS, etc.). LHC relies on them heavily (JetClass, Aspen Open Jets, ATLAS Top Tagging, ...). We desperately need benchmarks. Otherwise, we cannot compare our methods together, which means slowed scientific progress on methods.
- ...and is aligned with Genesis mission
- Please consider contributing your models to the DORAEMON open data challenge (LArTPC, WCh, soon IceCube, etc...) leaderboards (see [Kazu's talk tom](#)) come Q4 2026, or at least open source your method with reproducibility instructions and dataset (MicroBooNE open data, MINERvA, PILArNet, PILArNet-M).

It's worth it!



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2. Inefficiency

Point cloud learning is famously an unsolved field. PTv3 is SOTA, but is inefficient by the field's standards.

- ~60-70% of parameters come from the massive 3D sparse CNN kernels, with the rest going to the actual attention blocks. **Ratio of kernel to attn block params = $3^3 d^2 / 12 d^2 = 27/12 = 2.25$**
- I found that forcing the sparse CNN to operate in low dimensions even in late depths reduces parameters by half and keeps the same performance.
- See also [LitePT](#), [Volume Transformer](#) for very recent work on this issue.

Lessons learned, and challenges ahead

3. Scaling laws

- No understanding of predictive scaling laws in neutrino physics (see collider example [here](#))
- IMO, pre-training recipes should be general enough to be applied easily across similar problems in nu phys. We do not have this. Panda took ~4-5 months of tuning to work on LArTPC, ~1-2 week for WCh.

4. Parameter inefficient architectures.

Unlike natural images/language, point cloud learning is an “unsolved” field. Point Transformer v3 is SOTA, but is extremely inefficient.

For example:

- e.g., ~60-70% of parameters come from the massive 3D sparse CNN kernels, with the rest going to the actual attention blocks.
 - I found that forcing the sparse CNN to operate in low dimensions even in late depths reduces parameters by half and keeps the same performance.
- See also [LitePT](#) (arXiv:2512.13689), [Volume Transformer](#) (arXiv:2604.19609)

Summary and outlook

My site: <https://youngsm.com/>

Summary:

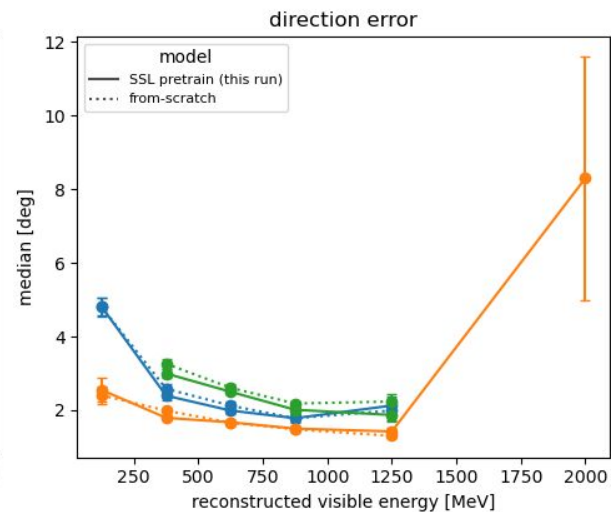
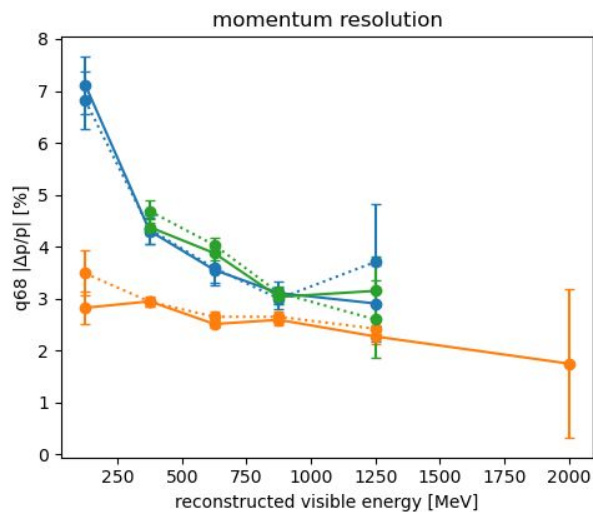
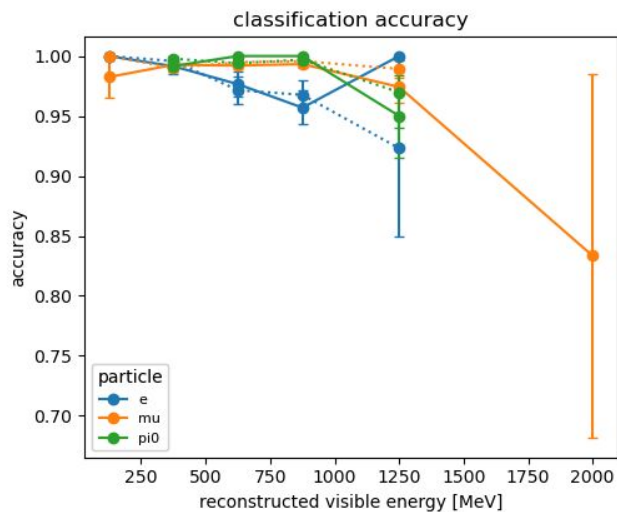
- Methods of label-free self-supervised learning in popular in computer vision can be applied successfully to HEP pattern recognition problems in different detection regimes (TPC, WCh) with minimal changes.
- The field is moving towards larger models, more data, and more compute.
 - High upfront cost of pretraining is largely amortized across downstream tasks.
 - The Genesis mission is also pushing heavily towards this...
- Efforts currently focus on efficient transfer learning across single-domain tasks, but...

Outlook:

- There is current interest to experiment with cross-experimental FMs (e.g., FM for DUNE ND+FD).
- Multi-modal pre-training being investigated (e.g., charge + light)
- Looking into seeing “FM”-ness by fine-tuning on other open datasets.
- AI-ready 200 TB dataset with O(10M) LArTPC + Water Cherenkov events targeted for Q4 2026. (See [Kazu's talk](#) and [Cesar's talk](#) on Friday)
- Please contribute (or create) easily reproducible common benchmarks on important tasks.

Extras

Function of E_{vis}



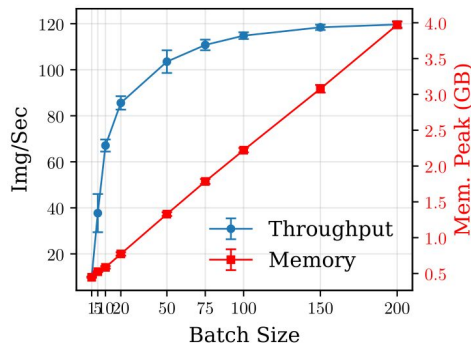
Needs!

Resources

- Compute: larger models, more compute, more data = better (FM scaling law)
 - Vision models are dominated largely by data memory (as opposed to a model)

Panda (A100)

Pre-training 10 epochs for
1M image dataset
(~ 10 GPU-day)



Config.	Batch	Params	Wall-time [s/iter]	Data Xfer [s/iter]	VRAM [GB]
Pre-train	12	104M	0.895	0.005	34.3
SemSeg (dec)	12	16.3M	0.210	0.001	3.4
PanSeg (vtx-dec)	6	4.3M	0.292	0.005	5.3
PanSeg (pid-dec)	6	4.3M	0.321	0.004	12.7

Needs!

Resources

- Compute: larger models, more compute, more data = better (FM scaling law)
 - Vision models are dominated largely by data memory (as opposed to a model)
 - A tiny R&D models 5-10 A100-day, for a large model x10-x100
 - Fine-tuning (task-specific models) ~ 1 A100-day per task
- Storage: large data size
 - PILArNet-1M (3D point cloud) ~167GB
 - PILArNet-10M (+ TPC/Optical waveforms) ~200TB
 - Publishing “encoded, AI-ready dataset” ... somewhere in between?

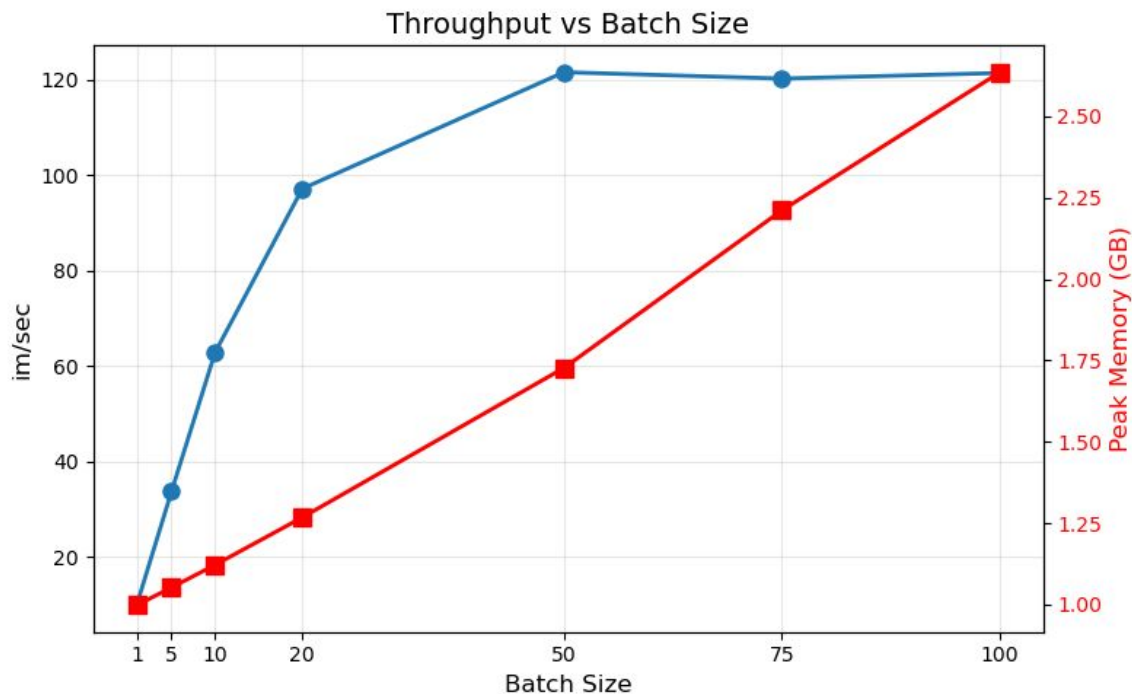
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 - Publishing “encoded, AI-ready dataset” ... somewhere in between?
- Software ecosystem - (always containers)
 - The core: pytorch, spconv, flash-attn, torch-scatter, pytorch-lightning, omegaconf (hydra), pybind11, hugging-face
 - Auxiliary: numba, cupy, jax, WandB, warpconvnet

Scalability

Semantic Segmentation (measured on single A100)

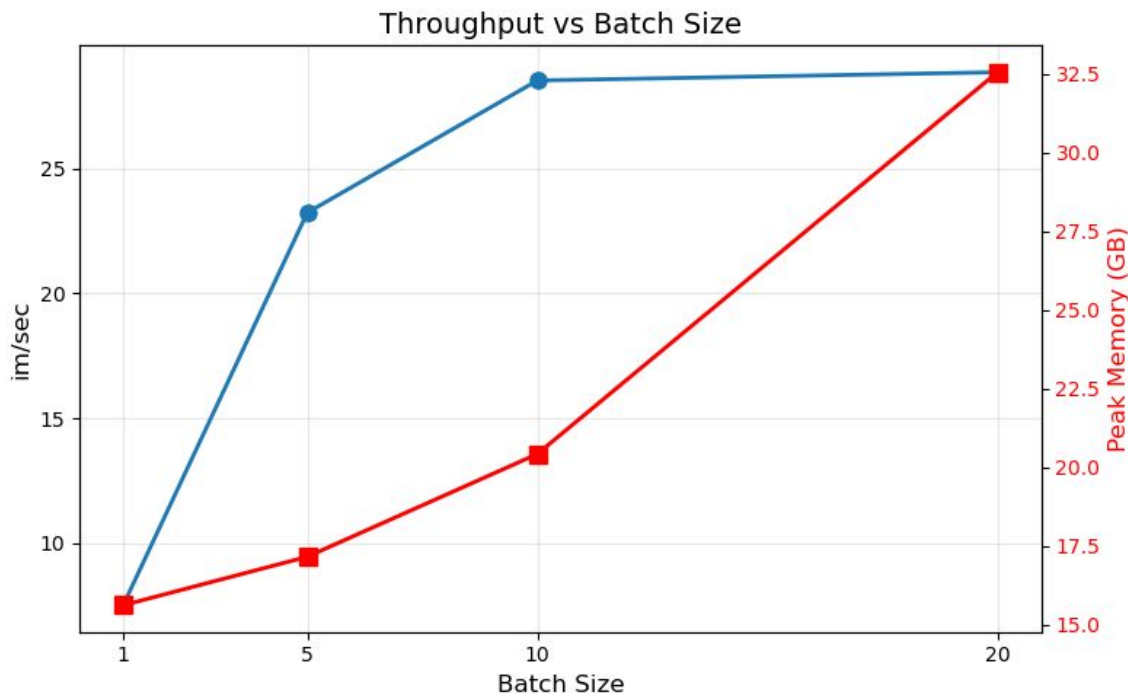


Peak throughput:
8.3 ms/image
120 img/sec

Mem@peak:
~1.75 GB

Scalability

Instance/Panoptic Segmentation (model forward + NMS post-processing)



Peak throughput:
34.5 ms / image
29 img/sec

Mem@peak:
~20 GB

NMS post-processing is
serial, but parallelized
via multiprocessing

Sharpening + Centering

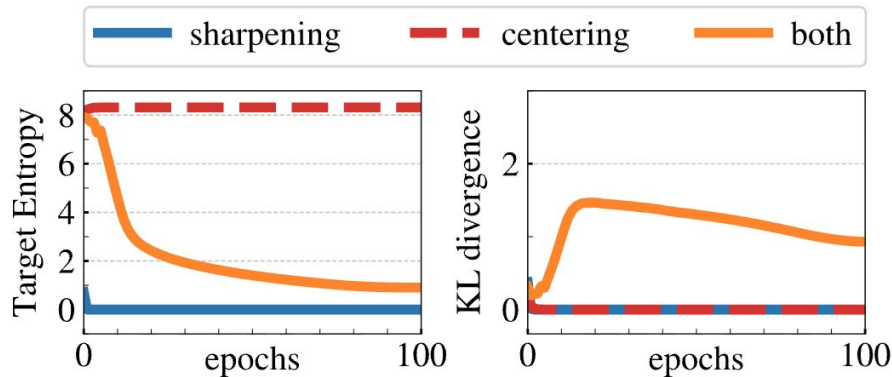


Figure 7: **Collapse study.** (left): evolution of the teacher’s target entropy along training epochs; (right): evolution of KL divergence between teacher and student outputs.

There are two forms of collapse: regardless of the input, the model output is uniform along all the dimensions or dominated by one dimension. The centering avoids the collapse induced by a dominant dimension, but encourages an uniform output. Sharpening induces the opposite effect. We show this complementarity by decomposing the cross-entropy H into an entropy h and the Kullback-Leibler divergence (“KL”) D_{KL} :

$$H(P_t, P_s) = h(P_t) + D_{KL}(P_t|P_s). \quad (5)$$

A KL equal to zero indicates a constant output, and hence a collapse.

Sharpening:

$$P_s(x)^{(i)} = \frac{\exp(g_{\theta_s}(x)^{(i)}/\tau_s)}{\sum_{k=1}^K \exp(g_{\theta_s}(x)^{(k)}/\tau_s)}, \quad (1)$$

with $\tau_s > 0$ a temperature parameter

Augmentations

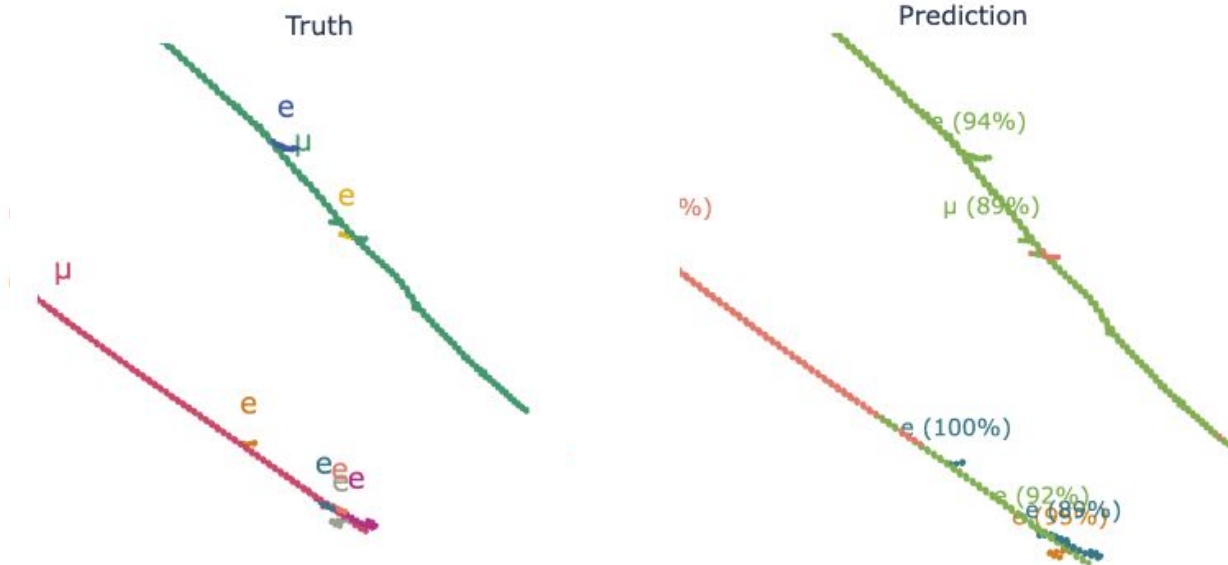
What about diffusion/attenuation?

Linear Evaluation on pre-training – 1M events



Poor instance reconstruction – particle

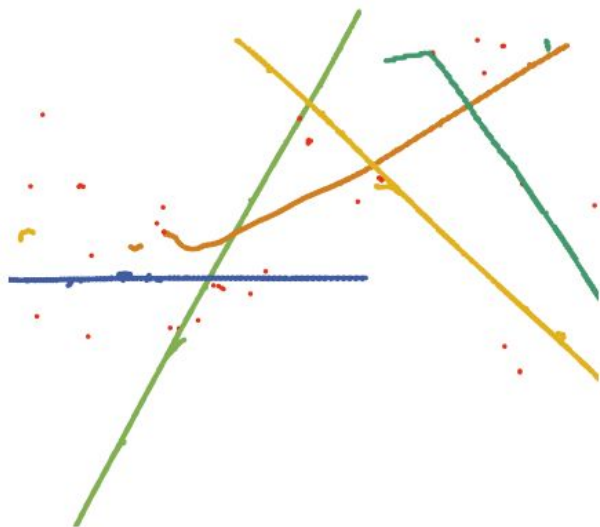
Instance Segmentation Comparison (ARI: 0.591)



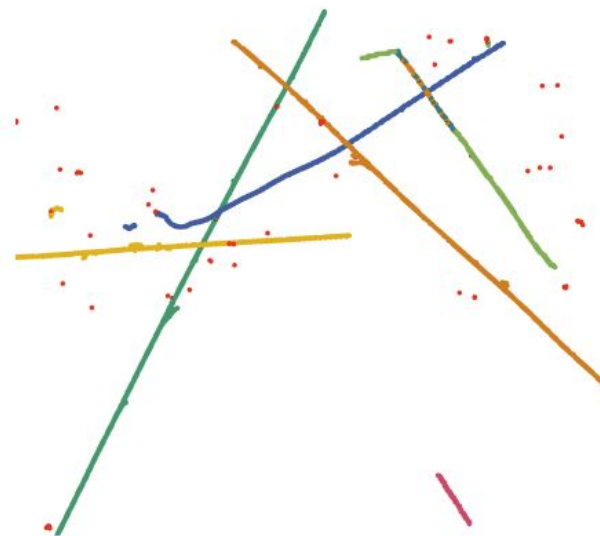
Poor instance reconstruction - interaction

Instance Segmentation Comparison (ARI: 0.941)

Truth

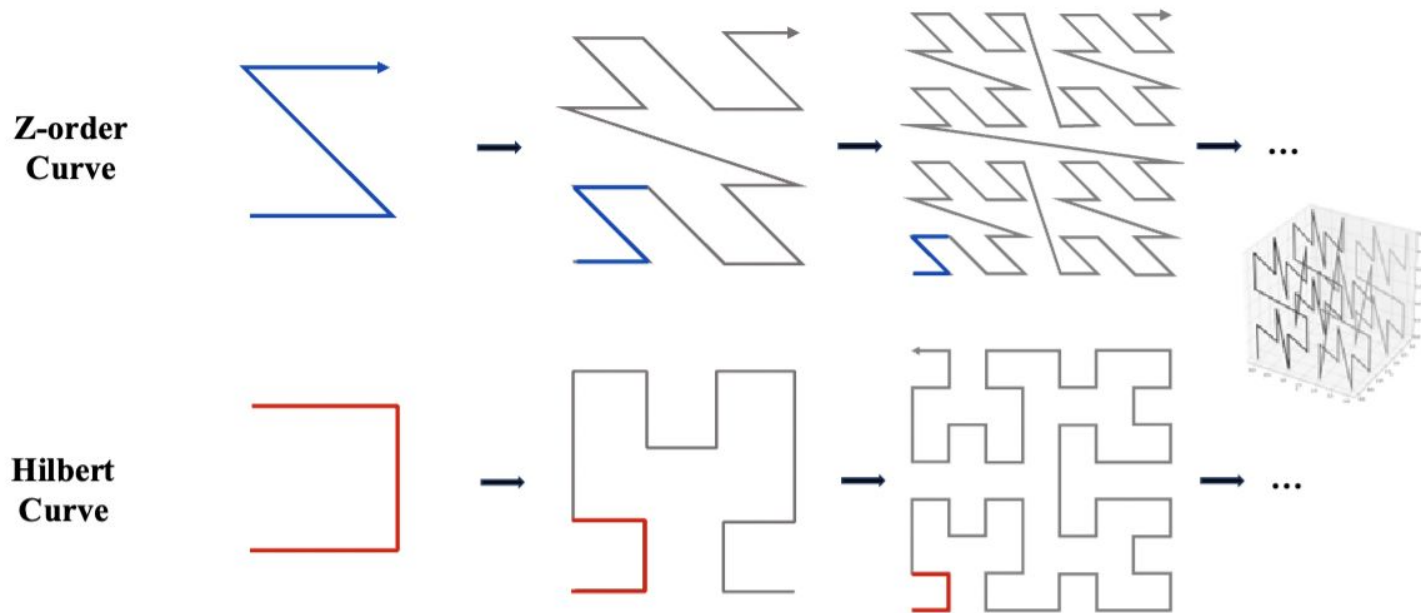


Prediction



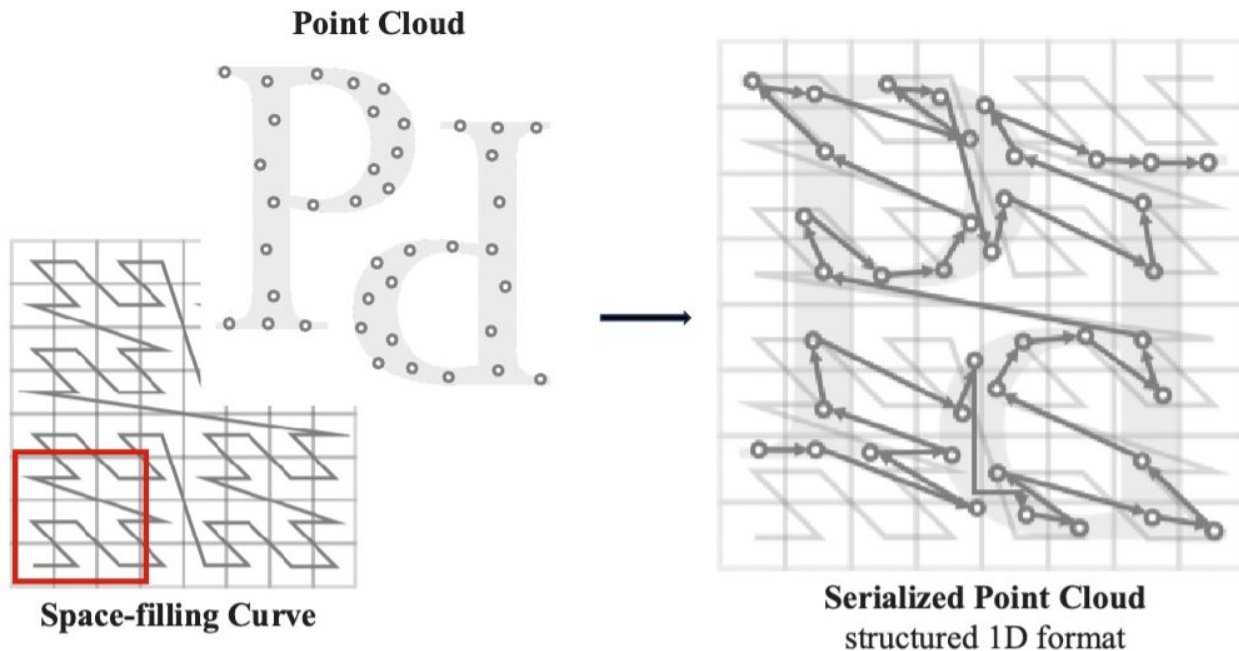
Serialized Attention

See Point Transformer V3 paper ([arXiv:2312.10035](https://arxiv.org/abs/2312.10035)) for more detail



Serialized Attention

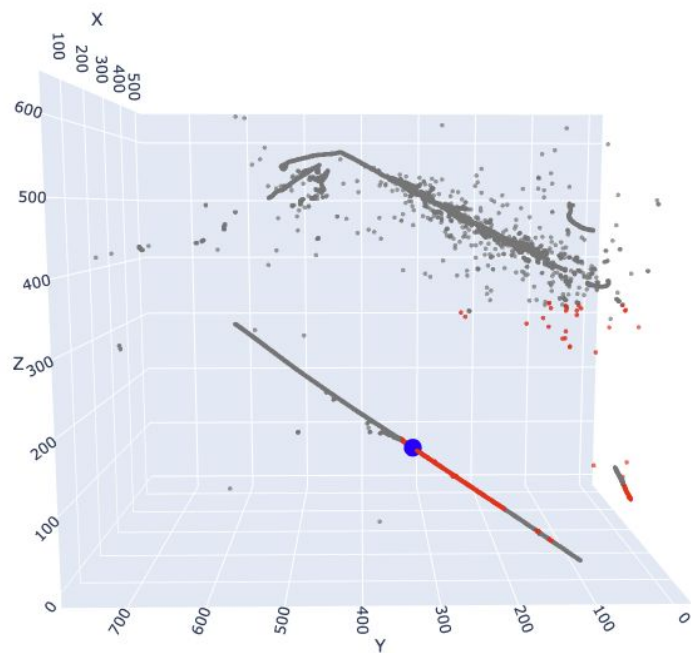
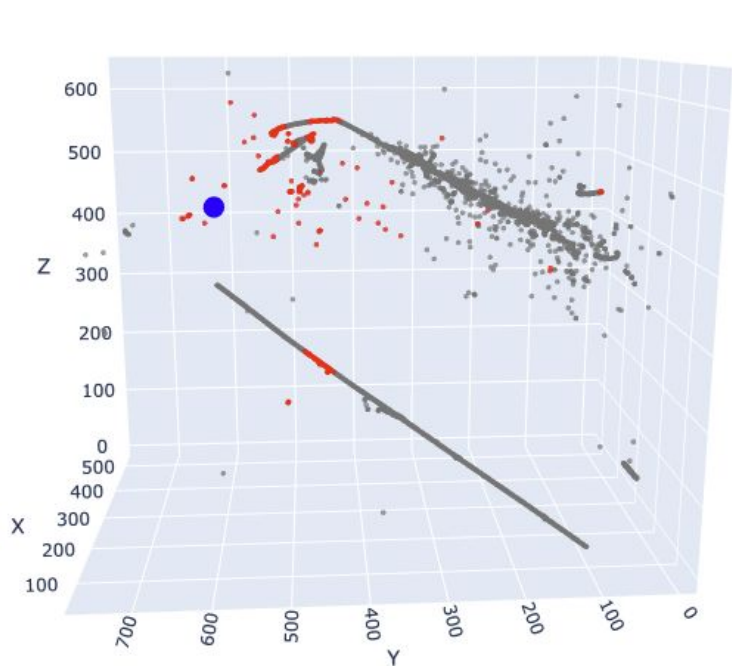
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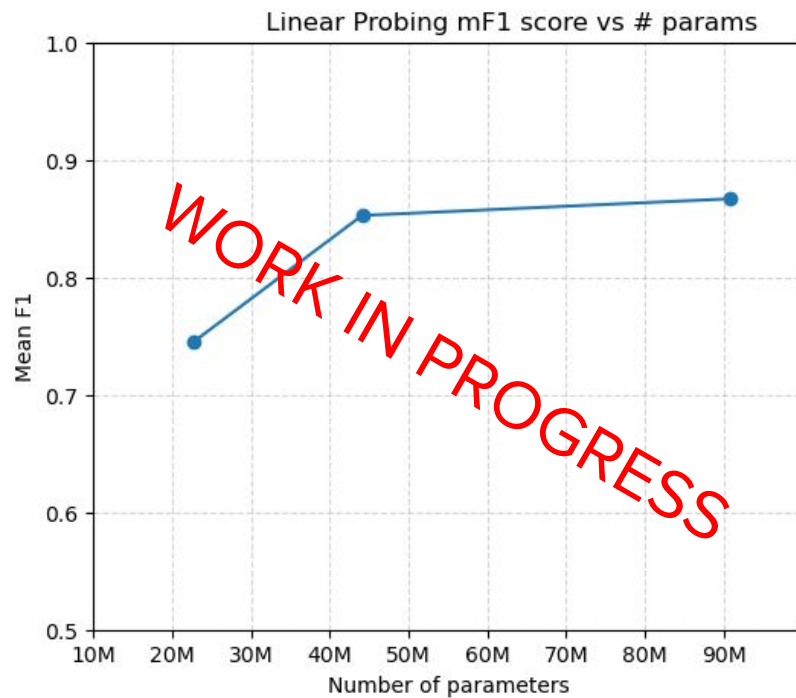
Serialized Attention – 256 voxel patches

Receptive field at a single point at stage 0

- not perfect, but good enough with enough depth.



Scaling Model Params



Pretraining techniques

INVARIANCE & RECONSTRUCTION

- **Masked reconstruction** $1D \cdot 2D \cdot 3D \cdot 1D+2D+3D$
Recover masked pixels, tokens, or local geometry from partial observations.

MAE, BEiT, iBOT, Point-BERT, Point-MAE, [PoLAR-MAE](#), Hiera / Point-M2AE, 4M
- **Non-contrastive alignment** $1D \cdot 2D \cdot 3D$
Teacher-student or twin-view agreement, usually without explicit negatives.

BYOL, SimSiam, DINO + DINOv2, Sonata, [Panda](#)
- **Predictive latent modeling** $2D$
Predict semantic latent targets rather than reconstructing raw pixels or prototype distributions
JEPA / LeJEPA / VICReg, data2vec, BYOL, point2vec
- **Contrastive alignment** $2D \cdot 3D$
Bring two views of the same sample together and push different samples apart.
SimCLR, MoCo, PointContrast, "Contrastive Learning for Robust Representations of Neutrino Data"

- Self-supervised learning
- Supervised learning

GENERATIVE, GEOMETRIC & CROSS-MODAL

- **Autoregressive token modeling** $2D$
Learn by next-token prediction over image tokens or rasterized sequences.

PointGPT, PointMamba, FM4NPP, NEPA
- **Generative: denoising / flow** $2D \cdot \text{multimodal}$
Learn representations through denoising, diffusion, or flow matching objectives.
Diffusion-based repr. learning, Self-Flow, VAE, "Score-based Diffusion Models for Generating LArTPC Images"
- **Geometry & multi-view 3D** $2D \rightarrow 3D \cdot 3D$
Use view consistency or scene geometry as the supervisory signal.

VGGT, Dust3r/Mast3r
- **Cross-modal alignment** $\text{any modality (+text)}$
Align representations across language, images, and point clouds.

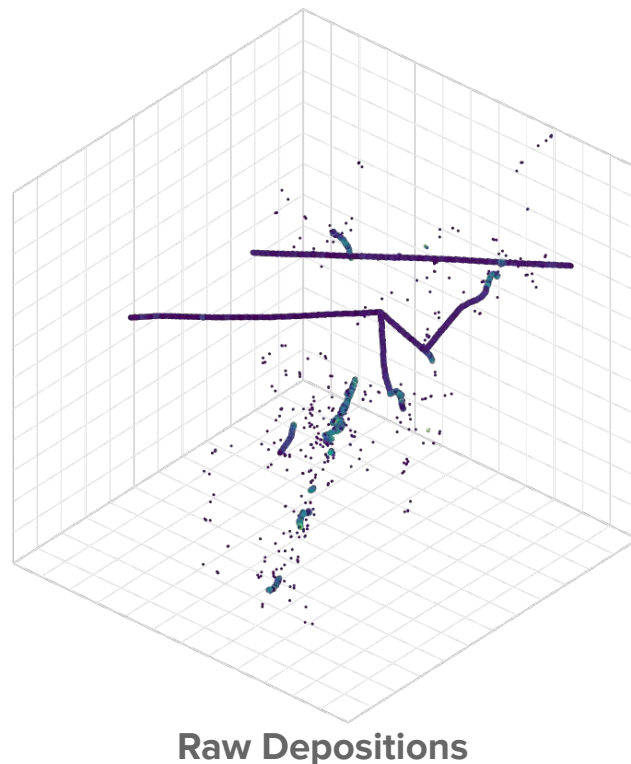
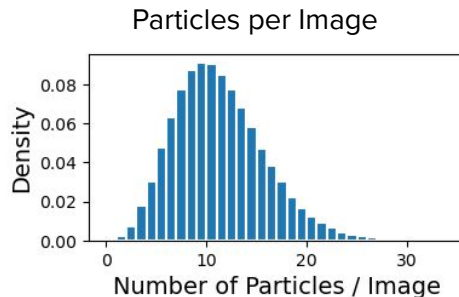
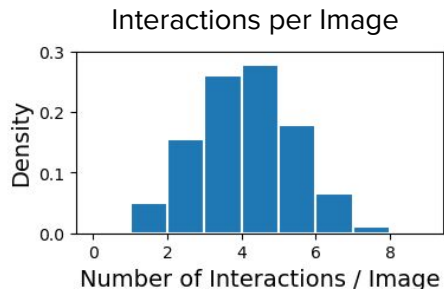
CLIP, Astro-CLIP, Perception Encoder, SigLIP, Concerto/Utonia
- **Labeled + multi-task supervision** supervised
General-purpose encoders trained on broad label spaces, segmentation masks, or task mixtures.
BiT / JFT, SAM, task mixtures, OmniLearn, L-GAtr

Dataset: PILArNet-Medium

Open data!

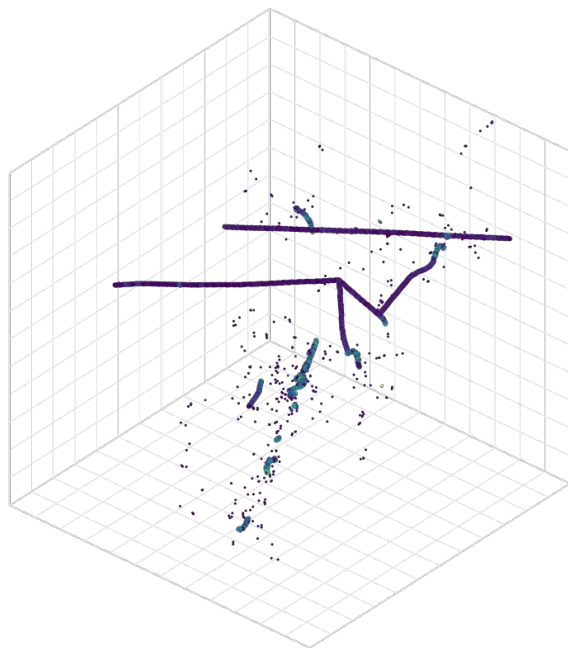


- Simulated dataset of 1.2M 3D events
- $(2.3 \text{ m})^3$ cube $(768 \text{ px})^3$. $\sim 5\text{B}$ non-zero voxels.
- +1M events on top of previous open dataset, [PILArNet \(2020\)](#).
- Simply 3D energy depositions, equivalent to “digital hits” from a LArTPC (e.g., DUNE Near Detector)
- 1024 - 30,000 voxels/event

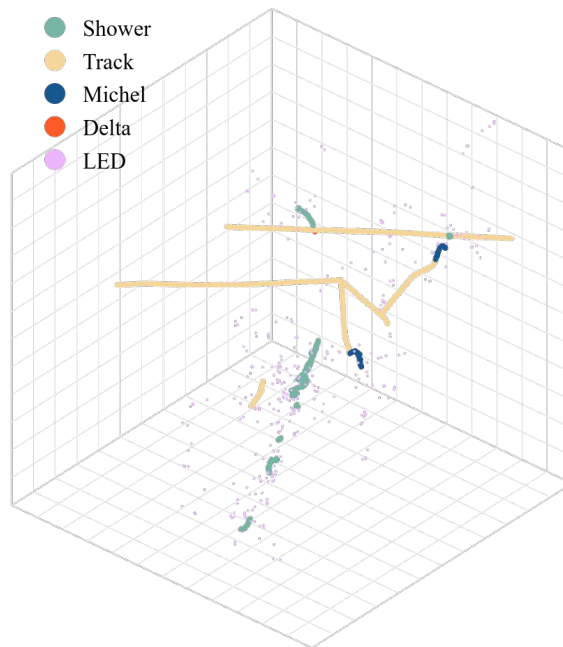


Semantic Segmentation Labels

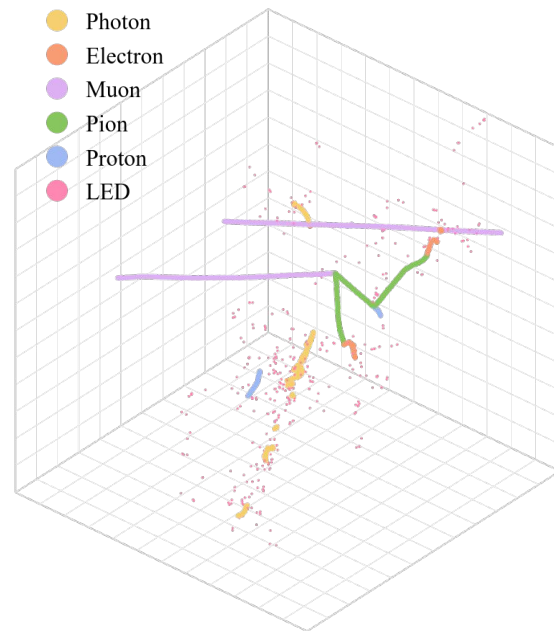
Open data!



Raw Depositions



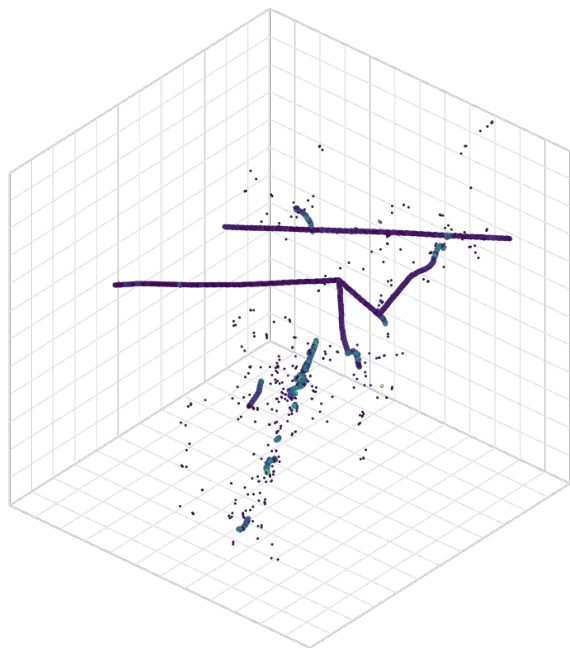
Semantics Reconstruction



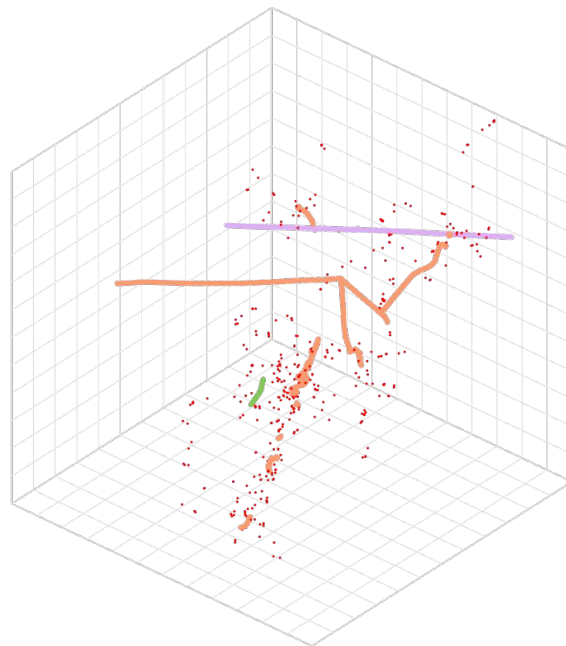
Particle ID Reconstruction

Instance Labels

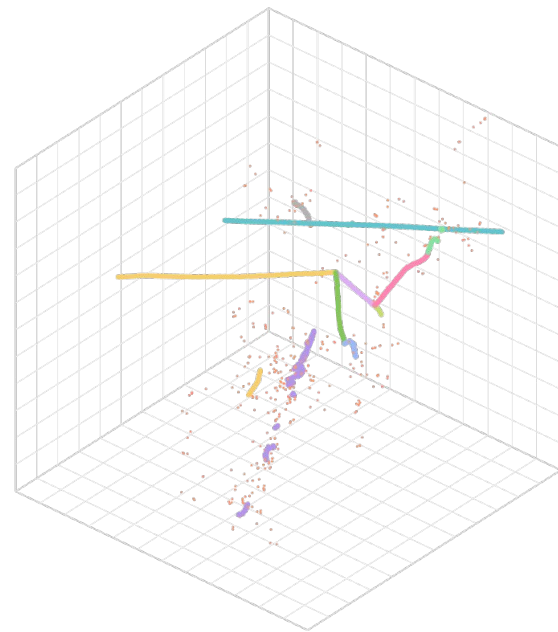
Open data!



Raw Depositions

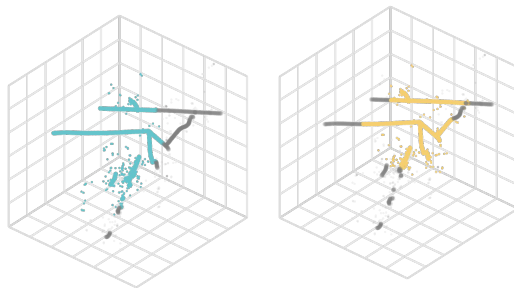


Interaction-level

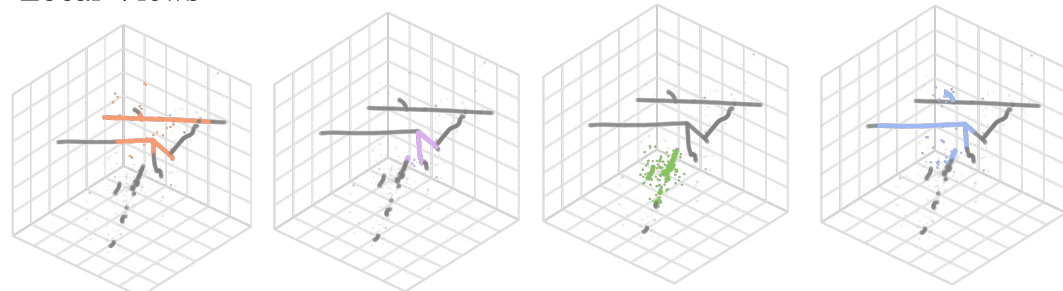


Particle-level

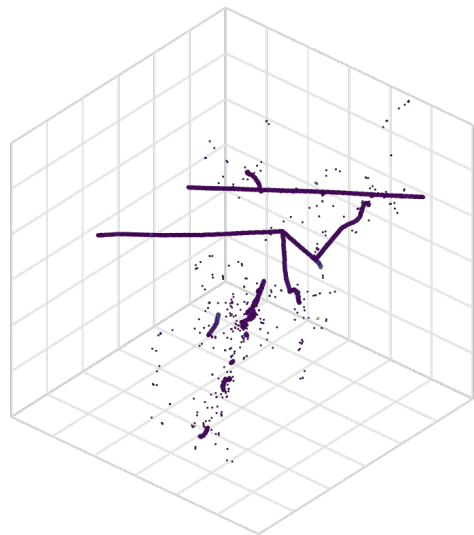
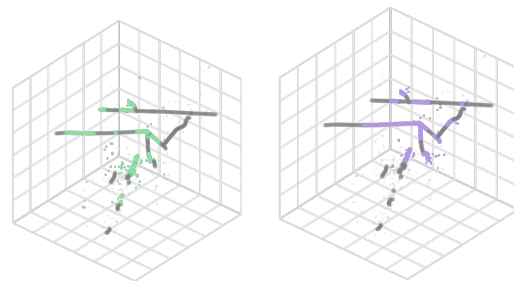
Global Views



Local Views



Masked Views



Full Image